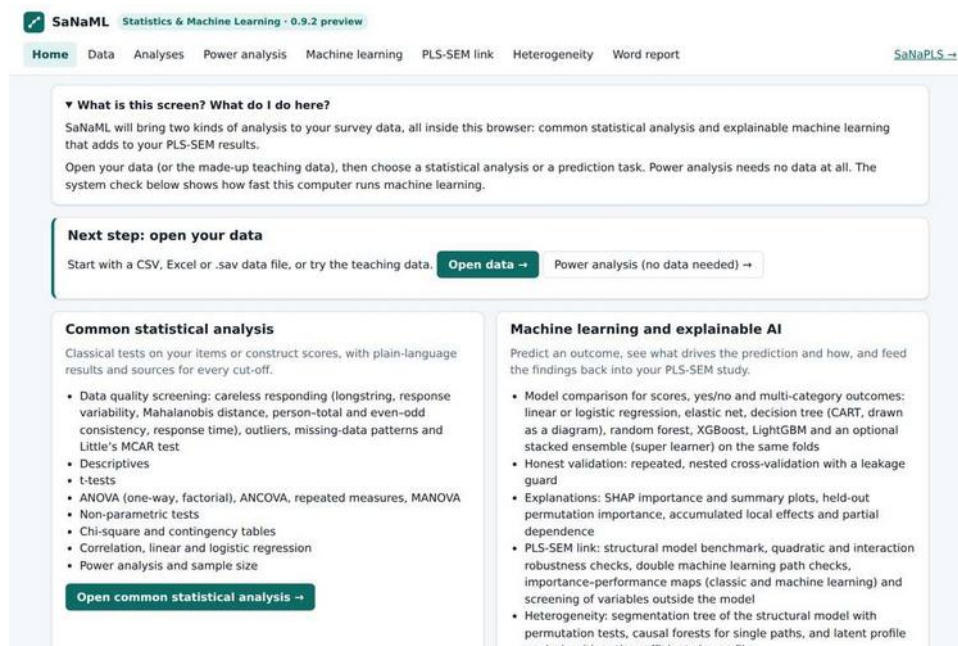


SaNaML

Statistics & Machine Learning: a step-by-step user manual

User manual • version 0.9.2 • October 2026



Worked examples: the built-in teaching data (240 made-up respondents), a careless-responding example and a UTAUT study with its SaNaPLS model

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About this manual

This manual shows you how to use SaNaML, the statistics and machine learning program of the SaNaSoft suite, step by step: from opening your data to downloading a Word report. Every chapter says what to click, what you will see, which rule of thumb applies, how to read the result and how to report it. All screenshots and numbers come from real runs of SaNaML 0.9.2, so you can follow along and get the same results.

Who this manual is for

Postgraduate students and researchers who analyse survey data. You do not need to know any statistics software. Each statistical idea is explained in plain language in a "Concept" box when it first appears; if you already know the idea, skip the box.

What you need

- A computer with an up-to-date browser (Google Chrome, Microsoft Edge, Mozilla Firefox or Safari). Machine learning runs inside the browser, so a recent computer helps; the system check in section 2.3 tells you how fast yours is.
- Your data as a CSV, Excel (.xlsx) or SPSS (.sav / .zsav) file with one row per respondent and one column per question, with the column names in the first row.
- For chapters 8 and 9: the results of your PLS-SEM model from SaNaPLS, saved as a .sanahandoff file (section 3.4).

How to read the instructions

Table 1

Conventions used in this manual

Convention	Meaning	Example
Bold green text	A button, tab, menu item, box or list on the screen. Click or choose exactly this.	Run the comparison
→	A sequence of clicks.	Data → Try the teaching data
Numbered steps	Actions to carry out in this order.	1. Open ... 2. Click ...
<i>Italics</i>	A technical term explained nearby, or text exactly as it appears in the results.	<i>nested cross-validation</i>

Concept: what a term means

Explains a statistical idea in plain language.

Cut-off values: the rule of thumb

The thresholds SaNaML applies, with their sources. SaNaML prints the same rule under every check ("Rule: ...").

What it means for our example

What the numbers of the worked example mean.

Tip

A shortcut or good practice.

Note

A warning or a point that needs judgement.

How to report it

A model sentence for your thesis or paper. SaNaML writes one for every result.

1. Introduction to SaNaML

1.1 What SaNaML does

SaNaML brings two kinds of analysis to your survey data:

- **Common statistical analysis:** data quality screening, descriptive statistics, t-tests, ANOVA (one-way, factorial, repeated measures), ANCOVA and MANOVA, non-parametric tests, chi-square and McNemar tests, correlation, linear and logistic regression, and power analysis.
- **Machine learning and explainable AI:** compare prediction models (from linear regression to XGBoost and LightGBM) on respondents they have not seen, explain what drives the predictions (SHAP, permutation importance, accumulated local effects), check calibration and precision, and use these tools to test and extend your PLS-SEM model (chapters 8 and 9).

Three principles apply throughout:

- **Your data stay on your computer.** Every calculation, machine learning included, runs inside your browser. Nothing is uploaded.
- **Plain language with sources.** Every result starts with *What it means*, lists its checks with a verdict (OK, Note, Check, Info) and the rule behind it, and ends with *How to report it (APA style)*.
- **Honest testing.** Prediction models are always judged on respondents they never saw, and every step that learns from the data is repeated inside each training set.

1.2 SaNaML in the SaNaSoft suite

SaNaML is one of three programs that work together. SaNaPLS estimates PLS-SEM models from questionnaire data; SaNaQual supports the thematic analysis of interviews; SaNaML adds classical statistics and machine learning, and can take over the construct scores and the model from SaNaPLS (section 3.4).

1.3 The worked examples

Table 2

The datasets used in this manual

Dataset	Where it comes from	Used in
Teaching data (made up), 240 respondents, 20 variables	Built in: Data → Try the teaching data	Chapters 3, 5, 6 and 7
careless_demo.csv, 300 respondents	A practice file with planted careless responders (satisfaction, trust, loyalty and value items, plus Duration in seconds)	Chapter 4
utaut_items_demo.csv and utaut.sanahandoff, 303 respondents	A UTAUT study (PE, EE, SI, FC, HM, BI, UB with Gender, AgeGroup and Experience) and its SaNaPLS model	Chapters 8 and 9

The teaching data contain ID, Gender (Female/Male), Education (Diploma/Bachelor/Postgraduate), Experience (years), the items PE1–PE4, EE1–EE3, EE4r (reverse-worded) and BI1–BI3 of a technology-adoption survey on a 1–7 scale, a satisfaction score measured three times (Satisfaction_Before, Satisfaction_After, Satisfaction_3Months) and whether the person uses the app at two points in time (Uses_App_T1, Uses_App_T2: No/Yes). The data are made up, so you can explore every analysis freely.

Tip: follow along

Open SaNaML next to this manual and repeat each step. Random parts of the analysis (cross-validation splits, bootstrap samples, permutations) use fixed seeds, so you get the same numbers as the screenshots. Machine learning timings depend on your computer; the results do not.

2. Getting started

2.1 Open SaNaML

- **In the browser:** go to sanapls.com and click **Statistics & Machine Learning →** on the landing page.
- **On your own computer (Windows):** unzip the SaNaSoft program and double-click **SaNaSoft.exe**, or double-click **Start SaNaSoft.bat** in the folder **ready-to-run**. Both open <http://localhost:4300/> in your browser; click **Statistics & Machine Learning →**. No internet connection is needed.

SaNaML opens on its Home screen (Figure 1).

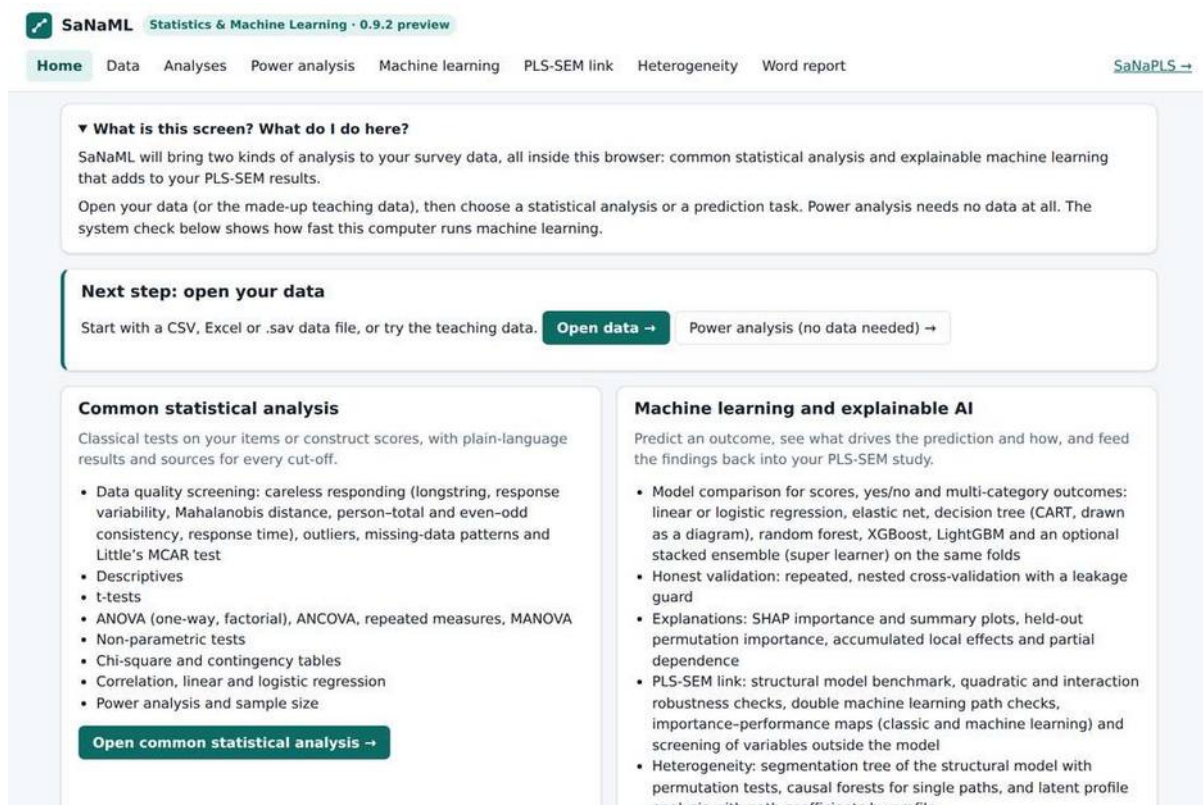


Figure 1. The Home screen of SaNaML.

2.2 The header and the Home screen

The header shows the program name with its version ("Statistics & Machine Learning · 0.9.2 preview") and the main menu: **Home**, **Data**, **Analyses**, **Power analysis**, **Machine learning**, **PLS-SEM link**, **Heterogeneity** and **Word report**. Once you have added results to the report, the last item shows their number, for example **Word report (5)**. **SaNaPLS →** on the right opens SaNaPLS.

Every screen starts with a box *What is this screen? What do I do here?*, which explains the screen in two or three sentences. Screens that need data show "No data yet" with a **Go to Data →** button until you open a file.

The Home screen has four cards: *Next step: open your data* (**Open data →** and **Power analysis (no data needed) →**), *Common statistical analysis*, *Machine learning and explainable AI* and *Is this computer ready for machine learning?*.

Note: where your work is kept

SaNaML keeps your data, results and report list in this browser and restores them on your next visit ("Restored from your last visit."). They are not visible in another browser or on another computer, and they are lost if you clear the browser's site data. Download the Word report (chapter 10) to keep your results.

2.3 The system check

Machine learning trains hundreds of models, so it is worth checking how fast your computer is before you start.

1. On **Home**, scroll to *Is this computer ready for machine learning?*.
2. Click **Check this computer**. SaNaML loads its model library, trains one real XGBoost model on made-up data and explains it with SHAP values.
3. Read the result (Figure 2).

Is this computer ready for machine learning?

SaNaML runs its models with WebAssembly, a fast, safe way for web pages to run compiled code, in background workers so the page stays responsive. The check trains one real XGBoost model on made-up data and explains it with SHAP values.

Yes

 WebAssembly

Yes

 WebAssembly SIMD (faster arithmetic)

Yes

 Background workers

2 Processor cores reported by the browser

Run the check again

Passed

 This computer is ready for SaNaML.

Model library	XGBoost 2.1.4, running in this browser
Time to load it	55 ms
One model (500 respondents, 6 predictors, 200 rounds)	229 ms
Predictions and SHAP values for everyone	201 ms
SHAP check (values add up to each prediction)	largest gap 5.7e-6 · passed
A full model comparison (500 models on 1 worker)	about 114 s

Figure 2. The system check: the browser features, the verdict and the timings.

The first list shows the browser features SaNaML needs: WebAssembly, WebAssembly SIMD (faster arithmetic) and background workers, and the number of processor cores. *Passed* means the computer is ready. The last row estimates how long a full model comparison takes on this computer (here about two minutes on one worker); more cores make it faster. If a required feature is missing, use a current version of Chrome, Edge, Firefox or Safari.

3. Bringing in your data

The **Data** screen has five numbered steps (Figure 3): 1 · *Open your data*, 2 · *Check the variable types*, 3 · *Build construct scores (optional)*, 4 · *Bring latent variable scores from SaNaPLS (optional)* and 5 · *Data quality screening*.

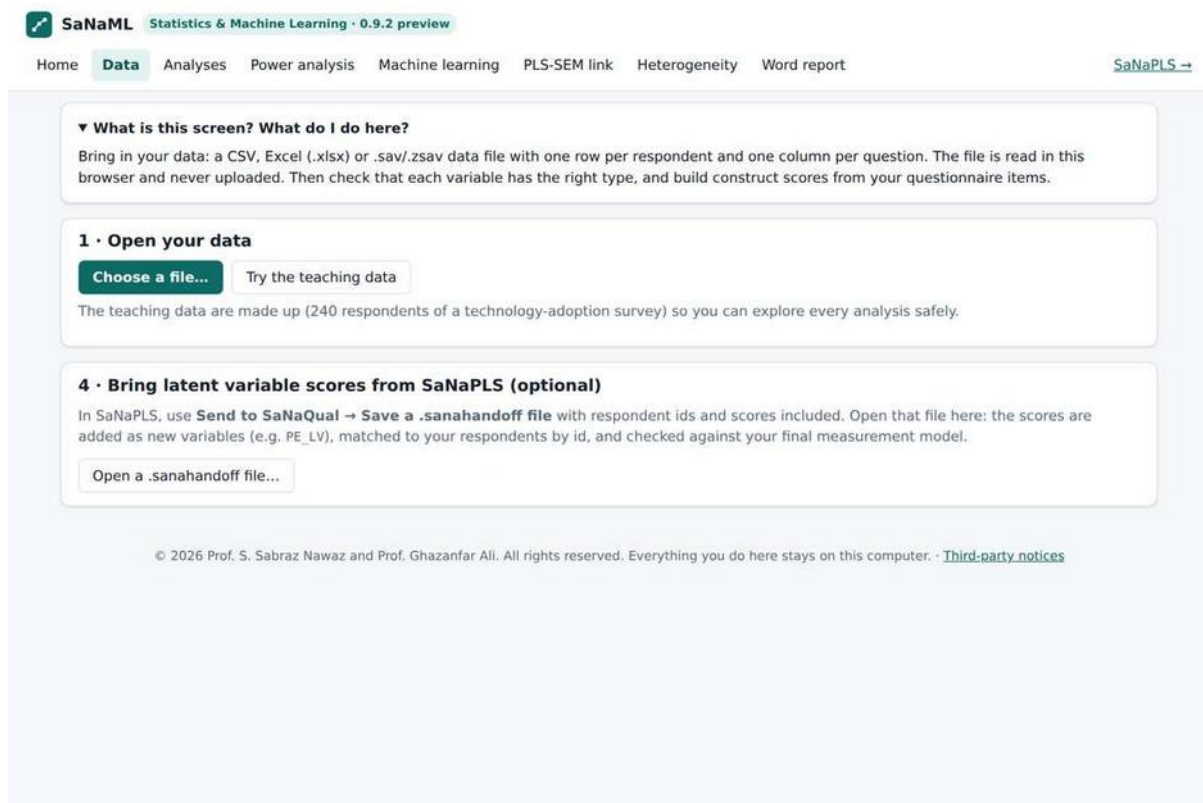


Figure 3. The Data screen before any data are open.

3.1 Step 1 · Open your data

1. Click **Data** in the menu.
2. For your own study, click **Choose a file...** and pick a CSV, Excel (.xlsx) or SPSS (.sav / .zsav) file. The file is read in the browser; it is never uploaded.
3. For this manual, click **Try the teaching data**. SaNaML reports "Open Teaching data (made up).csv: 240 rows, 20 variables."

Close this data closes the file again.

3.2 Step 2 · Check the variable types

SaNaML guesses the type of every variable (Figure 4):

- **Scale:** numbers you can average (ratings, scores, age, years of experience).
- **Categorical:** groups or answers without a meaningful average (gender, education, yes/no).

The table shows the number of valid and missing values, and the categories of each categorical variable. Change any type that is wrong in its list: the type decides which analyses offer the variable.

2 • Check the variable types

Scale = numbers you can average (ratings, scores, age). **Categorical** = groups or answers without a meaningful average (gender, education, yes/no). SaNaML guessed from your file; change any that are wrong.

Variable	Type	Valid	Missing	Values
ID	Scale	240	0	
Gender	Categorical	240	0	Female, Male
Education	Categorical	240	0	Diploma, Bachelor, Postgraduate
Experience	Scale	240	0	
PE1	Scale	240	0	
PE2	Scale	240	0	
PE3	Scale	240	0	
PE4	Scale	240	0	
EE1	Scale	240	0	
EE2	Scale	240	0	
EE3	Scale	240	0	
EE4r	Scale	240	0	
BI1	Scale	240	0	
BI2	Scale	240	0	
BI3	Scale	240	0	
Satisfaction_Before	Scale	240	0	
Satisfaction_After	Scale	240	0	
Satisfaction_3Months	Scale	240	0	
Uses_App_T1	Categorical	240	0	No, Yes
Uses_App_T2	Categorical	240	0	No, Yes

Figure 4. Step 2: the variable types of the teaching data.

What it means for our example

Gender, Education, Uses_App_T1 and Uses_App_T2 are categorical; all other variables are scale variables. No values are missing. ID is a scale variable here, but it is only a respondent number: SaNaML recognises such columns in machine learning and leaves them out (section 7.1).

3.3 Step 3 • Build construct scores

Most analyses in this manual work on construct scores, the average of the items that measure one idea (for example PE1–PE4 → PE). Step 3 builds them and checks first that the items hang together.

1. In **Name of the construct**, type **EE**.
2. In **Quick pick: items starting with**, type **EE** and press Tab. The items EE1, EE2, EE3 and EE4r are ticked.
3. EE4r is worded negatively, so tick its **R** box (reverse-coded). On a 1–7 scale, a reversed 2 becomes 6; **Lowest answer** and **Highest answer** (1 and 7) are filled in from the items.
4. Read Cronbach's α and the item statistics (Figure 5).

- Click **Create the construct score**. EE appears in the list of variables with a "construct" label.
- Build **PE** (PE1–PE4) and **BI** (BI1–BI3) in the same way.

3 • Build construct scores (optional)

A construct score is the average of the items that measure one idea (for example PE1–PE4 → PE). SaNaML first checks the items hang together (Cronbach's α). If you ran PLS-SEM in SaNaPLS, you will later be able to bring its latent variable scores instead.

Name of the construct
 Letters, numbers and underscores; must not be used by another variable yet.

Quick pick: items starting with

Lowest answer Highest answer
 Used to reverse-code items: on a 1–7 scale, a reversed 2 becomes 6.

Items (tick to include; "R" = reverse-coded)

- ☐ ID
- ☐ Experience
- ☐ PE1
- ☐ PE2
- ☐ PE3
- ☐ PE4
- ☒ EE1
- ☒ EE2
- ☒ EE3
- ☒ EE4r

☐ R
☐ R
☐ R
☒ R

$\alpha = .85$ Cronbach's α on 240 complete responses. Cronbach's α of .70 or higher is usually acceptable for research scales (Nunnally (1978)).

Item	Corrected item-total r	α if deleted
EE1	.67	.82
EE2	.68	.82
EE3	.71	.80
EE4r (R)	.71	.81

An item-total correlation below .30 (or a negative one) suggests the item does not fit, or that it needs reverse-coding.

Create the construct score

Figure 5. Step 3: building EE from four items, with EE4r reverse-coded.

Cut-off values: internal consistency

Cronbach's α $\geq .70$ is usually acceptable for research scales (Nunnally, 1978). The pill is green at .70 or more, amber from .60 and red below.

Corrected item-total correlation below .30, or negative, suggests that the item does not fit or needs reverse-coding.

What it means for our example

EE has $\alpha = .85$, and every corrected item-total correlation is between .67 and .71; no item would raise α if deleted. Without the R tick, EE4r would correlate negatively with the others: this is the usual sign of a reverse-worded item that was not reversed.

Note: construct scores and PLS-SEM

Simple averages weight every item equally. If you have estimated a PLS-SEM model in SaNaPLS, bring its latent variable scores instead (section 3.4): they use the weights of your measurement model.

3.4 Step 4 • Bring latent variable scores from SaNaPLS

The PLS-SEM link and heterogeneity screens (chapters 8 and 9) need the construct scores and the model from SaNaPLS.

1. In SaNaPLS, after the final bootstrapping run, click **Send to SaNaQual**, tick **Include respondent ids and scores**, and click **Save results for SaNaQual (.sanahandoff)**. (The same file serves SaNaQual and SaNaML.)
2. In SaNaML, open the questionnaire data of the same study with **Choose a file...** (here *utaut_items_demo.csv*, 303 rows).
3. In step 4, click **Open a .sanahandoff file...** and choose the file. SaNaML reports what it read: "7 constructs, 303 respondents with scores, with the measurement model" (Figure 6).
4. Check **Respondent id in your data** (the column SaNaPLS used as the id, here *RespID*) and choose **Which scores?: Unstandardised (on the items' scale, e.g. 1-7)** (recommended, easier to read in plots) or **Standardised (mean 0, SD 1)**.
5. Click **Add the scores** (Figure 7).

4 • Bring latent variable scores from SaNaPLS (optional)

In SaNaPLS, use **Send to SaNaQual** → **Save a .sanahandoff file** with respondent ids and scores included. Open that file here: the scores are added as new variables (e.g. *PE_LV*), matched to your respondents by id, and checked against your final measurement model.

Open a .sanahandoff file...

Read SaNaPLS 0.21.1, 7 constructs, 303 respondents with scores, with the measurement model.

Respondent id in your data **RespID** ▼

The column SaNaPLS used as the respondent id, so each score lands on the right respondent.

Which scores?

☒ **Unstandardised (on the items' scale, e.g. 1-7)**
Easier to read in plots and machine learning explanations ("above 5 on PE..."). Recommended.

☐ **Standardised (mean 0, SD 1)**
The scores SaNaPLS uses internally; useful for comparing effects across constructs.

Add the scores

Figure 6. Step 4 after opening the .sanahandoff file.

4 • Bring latent variable scores from SaNaPLS (optional)

In SaNaPLS, use **Send to SaNaQual** → **Save a .sanahandoff file** with respondent ids and scores included. Open that file here: the scores are added as new variables (e.g. *PE_LV*), matched to your respondents by id, and checked against your final measurement model.

Open a .sanahandoff file...

Read SaNaPLS 0.21.1, 7 constructs, 303 respondents with scores, with the measurement model.

Respondent id in your data **RespID** ▼

The column SaNaPLS used as the respondent id, so each score lands on the right respondent.

Which scores?

☒ **Unstandardised (on the items' scale, e.g. 1-7)**
Easier to read in plots and machine learning explanations ("above 5 on PE..."). Recommended.

☐ **Standardised (mean 0, SD 1)**
The scores SaNaPLS uses internally; useful for comparing effects across constructs.

Add the scores

Added *PE_LV, EE_LV, SI_LV, FC_LV, HM_LV, BI_LV, UB_LV* (unstandardised).

OK Same respondents. 303 of 303 respondents matched by *RespID*.

OK Same items. Every item of the final measurement model is in this file.

OK Same scores. Scores recomputed from the items and outer weights match the SaNaPLS scores ($r > .9999$ for every construct).

Figure 7. Step 4 after adding the scores: the new variables and three checks.

What it means for our example

The scores were added as PE_LV, EE_LV, SI_LV, FC_LV, HM_LV, BI_LV and UB_LV. All three checks are OK: all 303 respondents were matched by RespID, every item of the final measurement model is in the file, and the scores recomputed from the items and the outer weights match the SaNaPLS scores ($r > .9999$ for every construct). The data file and the SaNaPLS results therefore belong together.

3.5 Step 5 • Data quality screening

Check data quality → opens the screening of careless responders, outliers and missing answers. It is the subject of chapter 4. When your data are ready, **Choose an analysis →** at the bottom of the Data screen opens the analyses.

4. Data quality screening

Concept: careless responding

Some respondents answer without reading: the same answer everywhere, random answers, or far too fast. Their answers weaken correlations and can create or hide effects. No single index catches every kind of careless answering, so the usual advice is to compute several indices and exclude only respondents flagged by more than one (Curran, 2016; Meade & Craig, 2012).

In this chapter we use *careless_demo.csv* (300 respondents, 20 items: SAT1–5, TRU1–5, LOY1–5 and VAL1–5, plus Duration, the completion time in seconds).

1. On **Data**, open the file with **Choose a file...**, then click **Check data quality →**.
2. Check the *Screening settings* (Figure 8): the questionnaire items (all 20 are ticked), **Completion time in seconds** (SaNaML found **Duration**), **Suggest exclusion when flagged by (2 or more indices (Recommended))** and the **Longstring flag at** (by default half the items, at least 5).
3. Click **Screen the data**.

Screening settings (What to check)

Questionnaire items (in the order of the file)

Select all Select none

☒ SAT1	☒ SAT2	☒ SAT3	☒ SAT4	☒ SAT5	☒ TRU1	☒ TRU2
☒ TRU3	☒ TRU4	☒ TRU5	☒ LOY1	☒ LOY2	☒ LOY3	☒ LOY4
☒ LOY5	☒ VAL1	☒ VAL2	☒ VAL3	☒ VAL4	☒ VAL5	

Completion time in seconds: **Duration** Suggest exclusion when flagged by: **2 or more indices (Recommended)** Longstring flag at: **half the ite**

Reverse-coded items are taken from the construct scores you built; scales come from the constructs, the PLS-SEM model or item names (PE1, PE2 ... form PE).

Screen the data

Figure 8. The screening settings.

Table 3

The careless-response indices

Index	Flag when
Longstring (same answer many times in a row)	as many identical answers in a row as the longstring flag (here 10 of 20)
Intra-individual response variability (no variation at all)	the SD of the person's answers is 0
Mahalanobis distance (unusual answer pattern)	D^2 with $p < .001$
Person-total correlation (answers unlike everyone else's)	$r < 0$; used only when the item means differ enough (SD of the item means ≥ 0.25)
Even-odd consistency (inconsistent within scales)	Spearman-Brown corrected $r < .30$; needs three or more scales
Response time (too fast to have read the items)	under 2 seconds per answered item

4.1 Reading the screening report

Data quality screening (Careless responding, outliers and missing data)

What it means: Careless responding: 5 indices were computed on 20 items. 13 respondents were flagged by 2 or more indices and are suggested for exclusion (4.3% of 300); 27 more raised a single flag, which on its own is not enough to exclude anyone. Outliers: 0 values with $|z| > 3.29$ across 0 variables. Missing data: 1.7% of the answers are missing (at most 3.3% for one item); 217 respondents (72%) answered every item. Little's test does not reject that the data are missing completely at random ($\chi^2(648) = 562.58, p = .993$).

Note Careless responding. 13 respondents (4.3%) are suggested for exclusion. Look at their answers before deciding, and report the rule and the number excluded.

Rule: Use several careless-response indices and exclude only respondents flagged by more than one; report the indices, cut-offs and the number excluded (Curran (2016); Meade & Craig (2012)).

OK Outliers. No value is beyond $|z| = 3.29$.

Rule: Univariate outliers: $|z| > 3.29$ ($p < .001$); multivariate outliers: Mahalanobis D^2 with $p < .001$ (Tabachnick & Fidell (2019)).

OK Amount of missing data. At most 3.3% missing for one item.

Rule: Under 10% missing per variable is usually ignorable if the missingness is random; more needs a principled method (FIML or multiple imputation) (Hair et al. (2019)).

OK Missing completely at random. Little's test: $\chi^2(648) = 562.58, p = .993$. A non-significant result does not prove MCAR, but it gives no reason to doubt it.

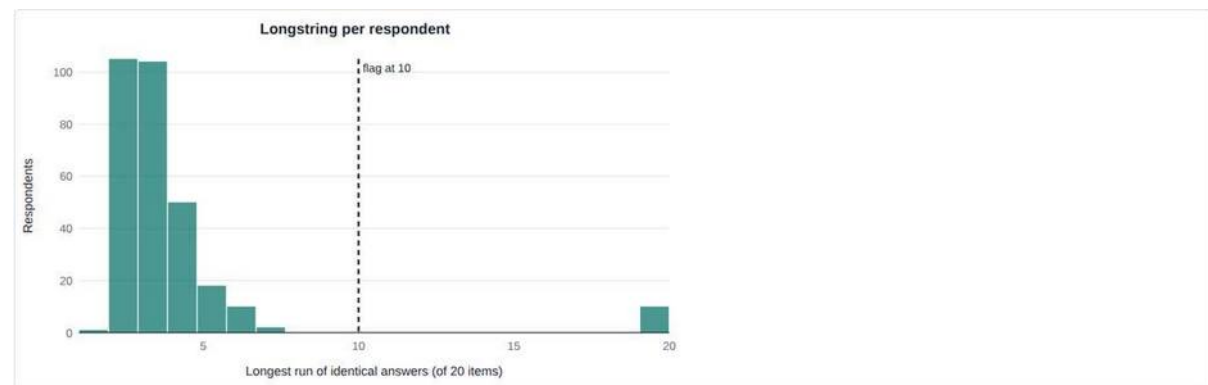
Rule: Little's test $p \geq .05$: no evidence against missing completely at random (Little (1988)).

Careless-response indices

Index	Flag when
Longstring (Same answer many times in a row)	10 or more identical answers in a row (of 20 items)
Intra-individual response variability (No variation at all)	SD of the answers = 0
Mahalanobis distance (Unusual answer pattern)	D^2 with $p < .001$ ($\chi^2, 20$ df; complete answers)
Person-total correlation (Answers unlike everyone else's)	Not used: the item means are too similar ($SD\ 0.05 < 0.25$) for the index to be informative
Even-odd consistency (Inconsistent within scales)	Spearman-Brown corrected $r < .30$ (4 scales)
Response time (Too fast to have read the items)	under 2 seconds per answered item

A respondent flagged by at least 2 indices is suggested for exclusion; no single index is decisive. Longstring follows the item order of the file. Reverse-coded items (SAT3, TRU3, LOY3, VAL3) were reversed for the person-total and even-odd indices; SAT3, TRU3, LOY3, VAL3 were not marked but correlate negatively with the rest of the items, so they were tested as reverse-coded. Factor: SAT (1), TRU (1), LOY (1), VAL (1).

Figure 9. The start of the screening report: what it means, the four checks and the indices used.



Longest run of identical answers per respondent.

Figure 10. Longest run of identical answers per respondent. The dashed line is the flag (10 of 20).

The report checks four things: careless responding, outliers ($|z| > 3.29$, and multivariate outliers by Mahalanobis D^2 with $p < .001$; Tabachnick & Fidell, 2019), the amount of missing data, and Little's MCAR test (Little, 1988). Below the checks are the table of indices, the respondents suggested for exclusion with their values on every index, the outliers, the missing answers by item and the most common patterns of missing answers, and three figures.

What it means for our example

Five indices were computed (the person-total correlation was not used, because the item means are too similar). Thirteen respondents (4.3% of 300) were flagged by two or more indices and are suggested for exclusion; 27 more raised a single flag, which on its own is not enough. There are no outliers. 1.7% of the answers are missing (at most 3.3% for one item), and Little's test does not reject that they are missing completely at random, $\chi^2(648) = 562.58$, $p = .993$.

4.2 Your decision

Nothing is deleted from your file: you decide. The card *Your decision (Which respondents to exclude)* lists every respondent with at least one flag (Figure 11); those suggested by the rule are ticked.

1. Look at the flagged respondents, and untick anyone you want to keep (or tick others).
2. Click **Add to Word report** under the screening report, so that the rule, the cut-offs and the number excluded are documented.
3. Click **Exclude the 13 ticked respondents**. The card *Screened data in use* confirms the exclusion (Figure 12): every analysis now uses the remaining 287 respondents.

Your decision (Which respondents to exclude)				
Respondents with at least one flag (40); ticked = to be excluded				
Exclude	Respondent	Flags	Longstring	IRV
☒	P001	Longstring, Intra-individual response variability	20	0.00
☒	P002	Longstring, Intra-individual response variability	20	0.00
☒	P003	Longstring, Intra-individual response variability	20	0.00
☒	P004	Longstring, Intra-individual response variability	20	0.00
☒	P005	Longstring, Intra-individual response variability	20	0.00
☒	P006	Longstring, Intra-individual response variability	20	0.00
☒	P007	Longstring, Intra-individual response variability	20	0.00
☒	P008	Longstring, Intra-individual response variability	20	0.00
☒	P009	Longstring, Intra-individual response variability	20	0.00
☒	P010	Longstring, Intra-individual response variability	20	0.00
Exclude the 13 ticked respondents				

Figure 11. Your decision: the flagged respondents, with those suggested for exclusion ticked.

Screened data in use
13 of 300 respondents are excluded: P001, P002, P003, P004, P005, P006, P007, P008, P009, P010, P013, P014, P019. Every analysis now uses the remaining 287.
Excluded after the data quality screening: flagged by at least 2 careless-response indices (or chosen by the researcher).
Undo the exclusion (use all 300 respondents)

Figure 12. Screened data in use, with the button to undo the exclusion.

Undo the exclusion (use all 300 respondents) brings everyone back.

Cut-off values: missing data

Under 10% missing per variable is usually ignorable if the missingness is random; more needs a principled method such as FIML or multiple imputation (Hair et al., 2019).

Little's MCAR test $p \geq .05$: no evidence against missing completely at random (Little, 1988). A significant test means the data are not MCAR, although they may still be missing at random.

How to report it

"Data were screened for careless responding with 5 indices (longstring, intra-individual response variability, Mahalanobis distance, even-odd consistency, response time; Curran, 2016; Meade & Craig, 2012); 13 of 300 respondents were flagged by at least 2 indices and excluded; 1.7% of item responses were missing, and Little's (1988) MCAR test was not significant, $\chi^2(648) = 562.58$, $p = .993$."

5. Common statistical analysis

Click **Analyses** in the menu. The menu on the left lists the analyses, each with a one-line description; the result appears on the right. We return to the teaching data, with the constructs EE, PE and BI built in section 3.3.

5.1 How every result is laid out

There is **no Run button**: the result appears as soon as the variables are chosen, and changes when you change them. Every result has the same parts:

- a title, for example *Independent-samples t-test (Comparing EE between Female and Male)*;
- **What it means**: the finding in one or two plain sentences;
- the **checks**, each with a verdict (OK, Note, Check or Info), an explanation and the rule with its source;
- the **tables** (and figures);
- **How to report it (APA style)**: a ready-made sentence, with **Copy this sentence** and **Add to Word report**;
- **References**, collapsed at the bottom.

Each test also computes its rank-based or exact alternative and tells you which one to read.

5.2 Help me choose

If you are not sure which test you need, click **Test selection guide (Help me choose)** (Figure 13).

1. Answer 1 · *What kind of answer are you analysing?*: a continuous outcome (a number or a rating) or a categorical outcome (a category).
2. Answer 2 · *What do you want to compare or test? (or What do you want to know?)*.
3. SaNaML shows the suggested test, for example **McNemar test**, and **Open it →** opens it with the right settings.

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Common statistical analysis

Test selection guide (Help me choose)
Answer two questions and SaNaML suggests the right test.

Descriptive statistics
Means, SDs, medians, skewness and kurtosis, overall or by group.

Independent-samples, paired-samples and one-sample t-tests (Compare two groups or two measurements)
Welch and Student t-tests, with the Mann-Whitney U and Wilcoxon signed-rank tests.

One-way ANOVA (Compare three or more groups)
Welch ANOVA, Tukey and Games-Howell post hoc tests, Kruskal-Wallis test with Dunn.

One-way repeated-measures ANOVA (Same people, three or more times)
Mauchly test with Greenhouse-Geisser and Huynh-Feldt corrections; Friedman test

Test selection guide (Help me choose)

1 · What kind of answer are you analysing?

☐ Continuous (interval or ratio) outcome (A number or a rating)
A Likert item, a construct score, age, income... something you can average.

☒ Categorical (nominal or ordinal) outcome (A category)
Gender, education level, yes/no, a choice among options.

2 · What do you want to know?

☐ Chi-square test of independence (Whether two categorical variables are related)
e.g. app use by gender.

☐ Chi-square goodness-of-fit test (Whether the categories are equally common)
e.g. even spread over education levels.

☒ McNemar test (Whether a yes/no answer changed)
Same people, same question, two times.

☐ Binary logistic regression (Which variables predict a yes/no answer)
e.g. what predicts using the app.

Suggested McNemar test **Open it →**

Each test also shows its rank-based or exact alternative and tells you which one to read, based on your data.

Figure 13. The test selection guide: a categorical outcome whose yes/no answer may have changed leads to the McNemar test.

5.3 Descriptive statistics

Choose **Descriptive statistics**, tick the scale variables to describe (here PE, EE and BI) and, optionally, a categorical variable in **Split by** (here Gender) (Figure 14). The table gives n, missing values, mean, SD, median, minimum, maximum, and bias-corrected skewness and kurtosis with their standard errors.

Descriptive statistics by Gender

What it means: No variable is strongly skewed (all $|\text{skewness}| \leq 1$).

Descriptive statistics

Variable	Gender	n	Missing	Mean	SD	Median	Min	Max	Skewness	Kurtosis
EE	Female	118	0	5.29	0.90	5.25	2.75	7.00	-0.35 (0.22)	0.03 (0.44)
EE	Male	122	0	5.29	0.96	5.50	2.75	7.00	-0.55 (0.22)	-0.02 (0.43)
PE	Female	118	0	4.49	0.88	4.50	2.75	6.50	0.07 (0.22)	-0.74 (0.44)
PE	Male	122	0	4.59	1.04	4.50	2.00	7.00	-0.09 (0.22)	-0.32 (0.43)
BI	Female	118	0	4.83	0.95	4.83	2.00	7.00	-0.27 (0.22)	0.01 (0.44)
BI	Male	122	0	4.94	0.94	5.00	2.67	7.00	-0.06 (0.22)	-0.31 (0.43)

Skewness and kurtosis are bias-corrected (G1, G2; kurtosis is excess kurtosis); standard errors in brackets. Median = weighted average at $(n + 1)p$.

How to report it (APA style): EE: $M = 5.29$, $SD = 0.93$, $n = 240$; PE: $M = 4.54$, $SD = 0.96$, $n = 240$; BI: $M = 4.89$, $SD = 0.95$, $n = 240$.

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Figure 14. Descriptive statistics of EE, PE and BI by gender.

What it means for our example

No variable is strongly skewed (all $|\text{skewness}| \leq 1$). Over all 240 respondents: EE $M = 5.29$, $SD = 0.93$; PE $M = 4.54$, $SD = 0.96$; BI $M = 4.89$, $SD = 0.95$.

5.4 t-tests: compare two groups or two measurements

Choose **Independent-samples**, **paired-samples** and **one-sample t-tests**. Under *What do you want to compare?* pick the design (Figure 15):

- **Independent-samples t-test:** two groups of different people. Choose the **Score to compare** and the **Groups**.
- **Paired-samples t-test:** the same people measured twice. Choose the first and the second measurement.
- **One-sample t-test:** one score against a fixed value (**Test value**, often the scale midpoint, 4 on a 1–7 scale).

Direction of the test is two-tailed by default; choose a one-tailed test only if your hypothesis predicted the direction before you saw the data.

Common statistical analysis

Test selection guide (Help me choose)
Answer two questions and SaNaML suggests the right test.

Descriptive statistics
Means, SDs, medians, skewness and kurtosis, overall or by group.

Independent-samples, paired-samples and one-sample t-tests (Compare two groups or two measurements)
Welch and Student t-tests, with the Mann-Whitney U and Wilcoxon signed-rank tests.

One-way ANOVA (Compare three or more groups)
Welch ANOVA, Tukey and Games-Howell post hoc tests, Kruskal-Wallis test with Dunn.

One-way repeated-measures ANOVA (Same people, three or more times)
Mauchly test with Greenhouse-Geisser and Huynh-Feldt corrections; Friedman test.

Factorial ANOVA, ANCOVA and MANOVA (Several

Independent-samples, paired-samples and one-sample t-tests (Compare two groups or two measurements)

What do you want to compare?

- ☒ **Independent-samples t-test (Two groups of different people)**
For example women vs men, or users vs non-users, on the same score.
- ☐ **Paired-samples t-test (The same people measured twice)**
For example satisfaction before and after, or two ratings by each respondent.
- ☐ **One-sample t-test (One score against a fixed value)**
For example: is the average agreement above the scale midpoint of 4?

Score to compare
EE

A scale variable: an item, a construct score or another number.

Groups
Gender (2 groups)

A categorical variable. If it has more than two groups, choose the two to compare below (or use One-way ANOVA).

Direction of the test

- ☒ **Two-tailed test (Any difference)**
The usual choice: a difference in either direction counts.
- ☐ **Right-tailed (one-tailed) test (Only "first is greater")**
Use only if your hypothesis predicted this direction before seeing the data.
- ☐ **Left-tailed (one-tailed) test (Only "first is smaller")**
Use only if your hypothesis predicted this direction before seeing the data.

Independent-samples t-test (Comparing EE between Female and Male)

What it means: Female and Male do not differ significantly on EE (difference 0.00; $d = 0.00$, negligible).

Info Which t-test. Welch's t-test is reported first because it is accurate whether or not the variances are

Figure 15. The t-test screen with the independent-samples design, EE compared between the genders.

Independent-samples t-test (Comparing EE between Female and Male)

What it means: Female and Male do not differ significantly on EE (difference 0.00; $d = 0.00$, negligible).

Info Which t-test. Welch's t-test is reported first because it is accurate whether or not the variances are equal.

Rule: Compare two means with Welch's t-test by default; it stays accurate when the variances differ (Delacre et al. (2017)).

OK Equal variances. Levene's test does not suggest unequal variances; Student's and Welch's tests agree closely.

Rule: Levene's test $p < .05$ suggests unequal variances; rely on Welch's test and Games-Howell comparisons (Levene (1960); Games & Howell (1976)).

OK Sample size. Every group has 30 or more cases, so the test of means is robust to non-normal data.
Rule: With about 30 or more cases per group, t-tests and ANOVA tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al. (2002)).

Info Effect size. $d = 0.00$ is negligible.

Rule: Cohen's d : about 0.2 small, 0.5 medium, 0.8 large (Cohen (1988)).

Group descriptives

Group	n	Mean	SD	Median	Skewness
Female	118	5.29	0.90	5.25	-0.35
Male	122	5.29	0.96	5.50	-0.55

t-tests (two-sided)

Test	t	df	p	Mean difference	95% CI
Welch's t (recommended)	0.03	237.82	.977	0.00	[-0.23, 0.24]
Student's t (equal variances)	0.03	238	.977	0.00	[-0.23, 0.24]

Effect size

Measure	Value	95% CI	Size
Cohen's d	0.00	[-0.25, 0.26]	negligible
Hedges' g	0.00		negligible

The CI of d inverts the noncentral t distribution (Steiger & Fouladi, 1997).

Rank-based alternative

Test	U	z	p	Rank-biserial r	Method
Mann-Whitney U	7083.0	-0.21	.831	-.02	normal approximation

Equal variances

Test	F	df	p
Levene (mean)	0.49	1, 238	.487
Brown-Forsythe (median)	0.40	1, 238	.527

How to report it (APA style): An independent-samples Welch's t-test showed that EE did not differ significantly between Female ($M = 5.29$, $SD = 0.90$) and Male ($M = 5.29$, $SD = 0.96$), $t(237.82) = 0.03$, $p = .977$, $d = 0.00$, 95% CI [-0.25, 0.26].

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Figure 16. The result of the independent-samples t-test.

The result (Figure 16) reports Welch's t-test first, because it is accurate whether or not the variances are equal (Delacre et al., 2017), followed by Student's t-test, the effect size (Cohen's d with its confidence interval, and Hedges' g), the Mann-Whitney U test as the rank-based alternative, and Levene's and the Brown-Forsythe tests of equal variances.

Cut-off values: t-tests

Welch's t-test by default (Delacre et al., 2017).

Sample size: with about 30 or more cases per group, t-tests tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al., 2002).

Cohen's d: about 0.2 small, 0.5 medium, 0.8 large (Cohen, 1988).

What it means for our example

Women ($M = 5.29$, $SD = 0.90$) and men ($M = 5.29$, $SD = 0.96$) do not differ on EE, $t(237.82) = 0.03$, $p = .977$, $d = 0.00$. With the paired design, satisfaction rose from before ($M = 4.35$, $SD = 0.89$) to after ($M = 4.82$, $SD = 1.20$), $t(239) = 9.40$, $p < .001$, $d_z = 0.61$, a medium change.

5.5 One-way ANOVA: compare three or more groups

Choose **One-way ANOVA**, the **Score to compare** (EE) and the **Groups** (Education) (Figure 17). The result gives the group descriptives, the ANOVA table, Welch's F with η^2 and ω^2 , pairwise comparisons (Tukey–Kramer, or Games–Howell when Levene's test shows unequal variances), the Kruskal–Wallis test with Dunn's pairwise comparisons (Holm-adjusted) as the rank-based alternative.

One-way ANOVA (Comparing EE across Education)

What it means: EE does not differ significantly across the 3 Education groups ($\omega^2 = -.007$).

OK Equal variances. Levene's test does not suggest unequal variances: the classic ANOVA and Tukey-Kramer comparisons are used.

Rule: Levene's test $p < .05$ suggests unequal variances: rely on Welch's test and Games-Howell comparisons (Levene (1960); Games & Howell (1976)).

OK Sample size. Every group has 30 or more cases, so the test of means is robust to non-normal data.

Rule: With about 30 or more cases per group, t-tests and ANOVA tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al. (2002)).

Info Effect size. $\omega^2 = -.007$ (negligible).

Rule: η^2 / ω^2 : about .01 small, .06 medium, .14 large (Cohen (1988)).

Group descriptives

Group	n	Mean	SD
Diploma	65	5.24	0.98
Bachelor	106	5.32	0.96
Postgraduate	69	5.29	0.84

One-way ANOVA

Source	SS	df	MS	F	p
Between groups	0.23	2	0.12	0.13	.874
Within groups	205.56	237	0.87		
Total	205.79	239			

Robust test and effect sizes

Measure	Value
Welch's F	$F(2, 143.88) = 0.12, p = .884$
η^2	.001
ω^2 (less biased)	-.007
Levene's test	$F(2, 237) = 1.39, p = .251$

Pairwise comparisons: Tukey-Kramer

Pair	Mean difference	95% CI	p (adjusted)
Diploma – Bachelor	–0.08	[–0.42, 0.27]	.862
Diploma – Postgraduate	–0.05	[–0.43, 0.33]	.946
Bachelor – Postgraduate	0.02	[–0.31, 0.36]	.984

Rank-based alternative

Test	H	df	p	ϵ^2
Kruskal-Wallis	0.25	2	.883	.001

Rank-based pairwise comparisons: Dunn's test (Holm-adjusted)

Pair	z	p	p (Holm)
Diploma – Bachelor	–0.38	.706	1.000
Diploma – Postgraduate	0.06	.955	1.000
Bachelor – Postgraduate	0.45	.655	1.000

How to report it (APA style): A one-way ANOVA showed no significant effect of Education on EE, $F(2, 237) = 0.13, p = .874, \eta^2 = .001, \omega^2 = -.007$.

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Figure 17. One-way ANOVA of EE across the three education groups.

Cut-off values: ANOVA effect sizes

η^2 and ω^2 : about .01 small, .06 medium, .14 large (Cohen, 1988). ω^2 is less biased in small samples.

Levene's test $p < .05$: rely on Welch's test and the Games-Howell comparisons (Levene, 1960; Games & Howell, 1976).

What it means for our example

EE does not differ across the education groups, $F(2, 237) = 0.13$, $p = .874$, $\eta^2 = .001$, $\omega^2 = -.007$.

5.6 Repeated-measures ANOVA: the same people three or more times

Choose **One-way repeated-measures ANOVA** and tick the measurements in the order they were taken (Satisfaction_Before, Satisfaction_After, Satisfaction_3Months) (Figure 18). Respondents with any missing measurement are left out. The result includes Mauchly's test of sphericity with the Greenhouse–Geisser and Huynh–Feldt corrections, pairwise paired t-tests (Holm-adjusted) and the Friedman test.

One-way repeated-measures ANOVA (Comparing Satisfaction_Before, Satisfaction_After, Satisfaction_3Months, same respondents)

What it means: The scores differ across Satisfaction_Before, Satisfaction_After, Satisfaction_3Months (small effect, generalised $\eta^2 = .028$). Pairs that differ: Satisfaction_Before vs Satisfaction_After; Satisfaction_Before vs Satisfaction_3Months; Satisfaction_After vs Satisfaction_3Months.

Note Sphericity. Mauchly's test is significant ($p < .001$); the Huynh-Feldt correction is used ($\epsilon = .763$).
Rule: If Mauchly's test is significant, correct the degrees of freedom: Greenhouse-Geisser when $\epsilon < .75$, Huynh-Feldt when $\epsilon \geq .75$ (Girden (1992)).

OK Sample size. 240 complete respondents: the ANOVA is robust to non-normal data.
Rule: With about 30 or more cases per group, t-tests and ANOVA tolerate non-normal data; with fewer, prefer the rank-based test if the data are clearly skewed (Lumley et al. (2002)).

Info Effect size. Generalised $\eta^2 = .028$ (small).
Rule: η^2 / ω^2 : about .01 small, .06 medium, .14 large (Cohen (1988)).

Descriptives (complete cases)				
Measure	n	Mean	SD	
Satisfaction_Before	240	4.35	0.89	
Satisfaction_After	240	4.82	1.20	
Satisfaction_3Months	240	4.66	1.31	

Repeated-measures ANOVA				
Correction	df	F	p	ϵ
None (sphericity assumed)	2, 478	25.78	< .001	1
Greenhouse-Geisser	1.52, 362.98	25.78	< .001	.759
Huynh-Feldt	1.53, 364.81	25.78	< .001	.763

Partial $\eta^2 = .097$; generalised $\eta^2 = .028$ (Bakeman, 2005).

Mauchly's test of sphericity			
W	χ^2	df	p
0.683	90.70	2	< .001

Pairwise paired t-tests (Holm-adjusted)						
Pair	Mean difference	t	df	p	p (Holm)	d_z
Satisfaction_Before – Satisfaction_After	–0.47	–9.40	239	< .001	< .001	–0.61
Satisfaction_Before – Satisfaction_3Months	–0.31	–4.86	239	< .001	< .001	–0.31
Satisfaction_After – Satisfaction_3Months	0.16	1.97	239	.050	.050	0.13

Rank-based alternative				
Test	χ^2	df	p	Kendall's W
Friedman	56.55	2	< .001	.118

Rank-based pairwise comparisons: Wilcoxon (Holm-adjusted)				
Pair	V	p	p (Holm)	r
Satisfaction_Before – Satisfaction_After	1148.0	< .001	< .001	–.74
Satisfaction_Before – Satisfaction_3Months	3754.5	< .001	< .001	–.39
Satisfaction_After – Satisfaction_3Months	8209.0	.046	.046	.17

How to report it (APA style): A one-way repeated-measures ANOVA with the Huynh-Feldt correction ($\epsilon = .763$) showed a significant difference across Satisfaction_Before, Satisfaction_After, Satisfaction_3Months, $F(1.53, 364.81) = 25.78$, $p < .001$, partial $\eta^2 = .097$, generalised $\eta^2 = .028$.

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Figure 18. Repeated-measures ANOVA of satisfaction at three points in time.

Cut-off values: sphericity

If Mauchly's test is significant, correct the degrees of freedom: Greenhouse-Geisser when $\epsilon < .75$, Huynh-Feldt when $\epsilon \geq .75$ (Girden, 1992).

What it means for our example

Mauchly's test is significant ($p < .001$) and $\epsilon = .763$, so the Huynh-Feldt correction is used. Satisfaction differs over time, $F(1.53, 364.81) = 25.78$, $p < .001$, partial $\eta^2 = .097$, generalised $\eta^2 = .028$ (small); all three pairs of time points differ.

5.7 Factorial ANOVA, ANCOVA and MANOVA

Choose **Factorial ANOVA, ANCOVA and MANOVA**. Under *What is your design?*:

- **Factorial ANOVA / ANCOVA**: one score, two or more grouping factors and/or covariates. Choose the **Score (outcome)**, tick the **Grouping factors**, and optionally **Covariates** (which makes it an ANCOVA). With two or more factors, keep the **Full factorial model** (with the interaction) unless you have a reason not to. The result uses Type III sums of squares and gives partial η^2 , estimated marginal means and Holm-adjusted pairwise comparisons.
- **One-way MANOVA**: several scores at once and one grouping factor. The result reports Pillai's trace (with Wilks, Hotelling-Lawley and Roy) and a follow-up ANOVA for each outcome (Olson, 1976).

What it means for our example

A factorial ANOVA of PE by Gender and Education shows an effect of Education, $F(2, 234) = 6.27$, $p = .002$, partial $\eta^2 = .051$, but no effect of Gender, $F(1, 234) = 1.19$, $p = .277$, and no interaction, $F(2, 234) = 2.27$, $p = .106$.

5.8 Chi-square, Fisher's exact and McNemar tests

Choose **Chi-square, Fisher's exact and McNemar tests**. Under *What do you want to know?*:

- **Chi-square test of independence**: are two categorical variables related? Fisher's exact test is added for small 2×2 tables. The result gives the counts with row percentages, adjusted standardised residuals and Cramér's V.
- **Chi-square goodness-of-fit test**: are the categories of one variable equally common?
- **McNemar test**: did the same yes/no answer change over time? (Figure 19)

McNemar test (Change from Uses_App_T1 to Uses_App_T2)

What it means: The share answering "Yes" went from 45% (Uses_App_T1) to 55% (Uses_App_T2): a significant change.

Info Which p-value. Enough respondents changed their answer for the chi-square version.

Paired counts

Uses_App_T1 \ Uses_App_T2	No	Yes
No	95	36
Yes	12	97

Tests

Test	Value	p
McNemar (continuity-corrected)	11.02	< .001
Exact binomial		< .001

How to report it (APA style): A McNemar test showed a significant change, $\chi^2(1) = 11.02$, $p < .001$.

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Figure 19. The McNemar test of app use at two points in time.

Cut-off values: chi-square

Expected counts: none below 1 and no more than 20% below 5; otherwise use the exact test (Cochran, 1954).

Cramér's V: about .10 small, .30 medium, .50 large (Cohen, 1988).

What it means for our example

Gender and app use at T1 are not related, $\chi^2(1, N = 240) = 2.92$, $p = .087$, Cramér's $V = .11$. App use rose from 45% at T1 to 55% at T2: 36 respondents started using the app and 12 stopped, a significant change, $\chi^2(1) = 11.02$, $p < .001$.

5.9 Correlation and regression

Choose **Correlation and regression analysis**. Under *What do you want to know?*:

- **Bivariate and partial correlation:** tick two or more scale variables and choose Pearson's r , Spearman's ρ or Kendall's τ -b. **Control for** makes it a partial correlation. p-values are Holm-adjusted (Figure 20).
- **Multiple linear regression:** choose the **Outcome (scale)**, the **Block 1 predictors** (for example control variables) and, optionally, the **Block 2 predictors** to see what they add (a hierarchical regression) (Figure 21). Categorical predictors, marked "(groups)", are dummy-coded against their first group.
- **Binary logistic regression:** choose an outcome with two groups, the answer that counts as the "event", and the predictors.

Pearson product-moment correlation (Pearson r)

What it means: 2 of 3 correlations are significant after Holm adjustment. The strongest is PE with BI ($r = .35$, medium, higher goes with higher).

Info Strength. Read the size of a correlation from its absolute value.

Rule: Correlation-type effect sizes (r , rank-biserial, ϕ): about .10 small, .30 medium, .50 large (Cohen (1988)).

Info Method. Pearson's r measures straight-line relationships. With strongly skewed data or many tied ratings, Spearman's ρ is a safer choice.

Correlation matrix

	1	2	3
1. EE	—		
2. PE	−.06	—	
3. BI	.32***	.35***	—

* $p < .05$, ** $p < .01$, *** $p < .001$ (Holm-adjusted). Each pair uses every respondent with both values.

Every pair

Pair	r	95% CI	n	p	p (Holm)
EE – PE	−.057	[−.18, .07]	240	.381	.381
EE – BI	.315	[.20, .43]	240	< .001	< .001
PE – BI	.350	[.23, .46]	240	< .001	< .001

How to report it (APA style): Pearson r correlations (Holm-adjusted p -values) showed: PE and BI, $r(238) = .35$, $p < .001$, 95% CI [.23, .46]; EE and BI, $r(238) = .32$, $p < .001$, 95% CI [.20, .43].

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Figure 20. Pearson correlations of EE, PE and BI.

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Common statistical analysis

Test selection guide (Help me choose)
 Answer two questions and SaNaML suggests the right test.

Descriptive statistics
 Means, SDs, medians, skewness and kurtosis, overall or by group.

Independent-samples, paired-samples and one-sample t-tests (Compare two groups or two measurements)
 Welch and Student t-tests, with the Mann-Whitney U and Wilcoxon signed-rank tests.

One-way ANOVA (Compare three or more groups)
 Welch ANOVA, Tukey and Games-Howell post hoc tests, Kruskal-Wallis test with Dunn.

One-way repeated-measures ANOVA (Same people, three or more times)
 Mauchly test with Greenhouse-Geisser and Huynh-Feldt corrections, Friedman test

Correlation and regression analysis (How variables go together and predict an outcome)

What do you want to know?

☐ **Bivariate and partial correlation (How strongly do variables go together?)**
 Correlations between two or more scores, optionally controlling for others.

☒ **Multiple linear regression (Which variables predict a score?)**
 For a scale outcome; add a second block to test what it adds (hierarchical regression).

☐ **Binary logistic regression (Which variables predict a yes/no outcome?)**
 For an outcome with two groups (e.g. uses the app: yes/no).

Outcome (scale)
 BI

The score you want to explain, e.g. a construct score.

Block 1 predictors (e.g. control variables)

☐ ID
 ☒ Gender (groups)
 ☐ Education (groups)
 ☒ Experience
 ☐ PE1
 ☐ PE2
 ☐ PE3
 ☐ PE4
 ☐ EE1
 ☐ EE2
 ☐ EE3
 ☐ EE4r
 ☐ BI1
 ☐ BI2
 ☐ BI3
 ☐ Satisfaction_Before
 ☐ Satisfaction_After
 ☐ Satisfaction_3Months
 ☐ Uses_App_T1 (groups)
 ☐ Uses_App_T2 (groups)
 ☐ EE
 ☐ PE

Block 2 predictors (optional: what do they add?)

☐ ID
 ☐ Education (groups)
 ☐ PE1
 ☐ PE2
 ☐ PE3
 ☐ PE4
 ☐ EE1
 ☐ EE2
 ☐ EE3
 ☐ EE4r
 ☐ BI1
 ☐ BI2
 ☐ BI3
 ☐ Satisfaction_Before
 ☐ Satisfaction_After
 ☐ Satisfaction_3Months
 ☐ Uses_App_T1 (groups)
 ☐ Uses_App_T2 (groups)
 ☒ EE
 ☒ PE

Multiple linear regression (Predicting BI)

What it means: The predictors together explain 23.7% of the variance in BI ($R^2 = .24$, $f^2 = 0.31$, medium). Significant predictors: EE ($\beta = .34$), PE ($\beta = .37$). Block 2 adds $\Delta R^2 = .234$ ($p < .001$).

Figure 21. Multiple linear regression of BI in two blocks: Gender and Experience, then EE and PE.

Multiple linear regression (Predicting BI)

What it means: The predictors together explain 23.7% of the variance in BI ($R^2 = .24$, $f^2 = 0.31$, medium). Significant predictors: EE ($\beta = .34$), PE ($\beta = .37$). Block 2 adds $\Delta R^2 = .234$ ($p < .001$).

OK Sample size. 240 respondents for 4 predictor terms (needed: 82).

Rule: At least $50 + 8k$ respondents for k predictors to test the whole model (Green (1991)).

OK Collinearity. Largest VIF = 1.05.

Rule: VIF below 5: no concern; 5–10: moderate; above 10: serious collinearity (Hair et al. (2019); Kutner et al. (2005)).

OK Normal residuals. Shapiro-Wilk $W = 0.993$, $p = .382$.

Rule: Shapiro-Wilk $p \geq .05$ for the residuals; with about 300 or more respondents the coefficient tests are robust anyway (Shapiro & Wilk (1965)).

Note Constant error variance. Breusch-Pagan (Koenker) $\chi^2(4) = 11.90$, $p = .018$.

Rule: Breusch-Pagan (Koenker) $p \geq .05$: no evidence that the error spread changes (Breusch & Pagan (1979); Koenker (1981)).

OK Influential cases. Largest Cook's distance = 0.088; 1 standardised residual beyond ± 3 .

Rule: No Cook's distance above 1 (Cook & Weisberg (1982)).

Info Independent errors. Durbin-Watson = 2.07 (values near 2 mean no order effect; it matters only when the row order means something, such as time).

Info Effect size. $f^2 = 0.31$ (medium).

Rule: f^2 : .02 small, .15 medium, .35 large (Cohen (1988)).

Model summary

Model	Predictors	R^2	Adjusted R^2	F	df	p	ΔR^2	p (ΔR^2)
1	Gender, Experience	.003	-.005	0.38	2, 237	.684		
2	Gender, Experience, EE, PE	.237	.224	18.25	4, 235	< .001	.234	< .001

Coefficients (model 2)

Predictor	B	SE	β	t	p	95% CI of B	VIF
(Intercept)	1.417	0.419		3.38	< .001	[0.592, 2.242]	
Gender: Male (vs Female)	0.071	0.108	.037	0.65	.513	[-0.142, 0.283]	1.01
Experience	-0.003	0.012	-.017	-0.28	.778	[-0.026, 0.020]	1.05
EE	0.346	0.059	.339	5.84	< .001	[0.229, 0.463]	1.04
PE	0.359	0.056	.366	6.38	< .001	[0.248, 0.470]	1.01

$n = 240$. Categorical predictors are dummy-coded against their first group. β = standardised coefficient.

How to report it (APA style): A multiple linear regression (two blocks) of BI on Gender, Experience, EE, PE was significant, $F(4, 235) = 18.25$, $p < .001$, $R^2 = .24$, adjusted $R^2 = .22$. EE predicted BI, $\beta = .34$, $t(235) = 5.84$, $p < .001$; PE predicted BI, $\beta = .37$, $t(235) = 6.38$, $p < .001$.

Copy this sentence

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► References

Figure 22. The regression result: checks, model summary and coefficients.

The regression result (Figure 22) checks the sample size, collinearity (VIF), normality of the residuals, constant error variance (Breusch-Pagan), influential cases (Cook's distance) and independent errors (Durbin-Watson), then gives the model summary (R^2 , adjusted R^2 , ΔR^2) and the coefficients (B, SE, β , t, p, the confidence interval and VIF).

Cut-off values: regression

Sample size: at least $50 + 8k$ respondents for k predictors to test the whole model (Green, 1991).

VIF: below 5 no concern; 5–10 moderate; above 10 serious (Hair et al., 2019).

Logistic regression: at least 10 respondents in the smaller outcome group per coefficient (EPV $\geq$ 10; Peduzzi et al., 1996); AUC .70 acceptable, .80 excellent, .90 outstanding (Hosmer et al., 2013).

What it means for our example

PE and BI correlate $r = .35$ and EE and BI $r = .32$ (both $p < .001$); EE and PE are unrelated ($r = -.06$). In the regression, Gender and Experience explain almost nothing ($R^2 = .003$); adding EE and PE raises R^2 by .234 ($p < .001$) to .24, $F(4, 235) = 18.25$, $p < .001$. EE ($\beta = .34$) and PE ($\beta = .37$) both predict BI.

How to report it

"A multiple linear regression (two blocks) of BI on Gender, Experience, EE, PE was significant, $F(4, 235) = 18.25$, $p < .001$, $R^2 = .24$, adjusted $R^2 = .22$. EE predicted BI, $\beta = .34$, $t(235) = 5.84$, $p < .001$; PE predicted BI, $\beta = .37$, $t(235) = 6.38$, $p < .001$."

6. Power analysis and sample size

Power analysis needs no data. Click **Power analysis** in the menu (Figure 23).

1. 1 • Which analysis will you run? Choose the analysis, for example **Multiple linear regression, F-test of R^2 (also the usual PLS-SEM check)**. For PLS-SEM, use the largest number of arrows pointing at one construct as the number of predictors.
2. 2 • What do you want to find? **A priori power analysis (The sample size I need)** before data collection, **Post hoc power analysis** for a given sample, or **Sensitivity power analysis (The smallest effect I can detect)**.
3. Enter the effect size, the significance level α (.05), the power (.80) and the details of the design (here 4 predictors).
4. Read the answer and copy the sentence under *How to report it*.

SaNaML Statistics & Machine Learning · 0.9.2 preview

Home Data Analyses **Power analysis** Machine learning PLS-SEM link Heterogeneity Word report (4) [SaNaPLS →](#)

▼ What is this screen? What do I do here?

Work out how many respondents you need *before* collecting data (a priori), how much power a planned or finished study has, or the smallest effect it can detect. No data file is needed. The sentence at the bottom can go straight into your proposal or methods section.

1 • Which analysis will you run?

- ☒ **Multiple linear regression, F-test of R^2 (also the usual PLS-SEM check)**
For PLS-SEM, use the largest number of arrows pointing at one construct as the number of predictors.
- ☐ **Inverse square root method (PLS-SEM minimum sample size)**
Minimum sample size from the smallest path coefficient you expect (Kock & Hadaya, 2018).
- ☐ **Independent-samples t-test (Compare two groups)**
Two groups of different people.
- ☐ **Paired-samples t-test (Same people twice)**
Before vs after, or two ratings per person.
- ☐ **One-sample t-test (One score against a fixed value)**
Compares one mean with a fixed value.
- ☐ **One-way ANOVA, fixed effects (Compare three or more groups)**
One-way ANOVA, effect size f .
- ☐ **Chi-square test (Goodness of fit or independence)**
Goodness of fit or independence; effect size w .
- ☐ **Pearson correlation test (Is a correlation different from zero?)**
Test whether a correlation differs from zero.

2 • What do you want to find?

- ☒ **A priori power analysis (The sample size I need)**
Most common in proposals.

Figure 23. An *a priori* power analysis for a regression with four predictors.

▼ What is this screen? What do I do here?

Work out how many respondents you need *before* collecting data (a priori), how much power a planned or finished study has, or the smallest effect it can detect. No data file is needed. The sentence at the bottom can go straight into your proposal or methods section.

1 • Which analysis will you run?

☐ Multiple linear regression, F-test of R^2 (also the usual PLS-SEM check)

For PLS-SEM, use the largest number of arrows pointing at one construct as the number of predictors.

☒ Inverse square root method (PLS-SEM minimum sample size)

Minimum sample size from the smallest path coefficient you expect (Kock & Hadaya, 2018).

☐ Independent-samples t-test (Compare two groups)

Two groups of different people.

☐ Paired-samples t-test (Same people twice)

Before vs after, or two ratings per person.

☐ One-sample t-test (One score against a fixed value)

Compares one mean with a fixed value.

☐ One-way ANOVA, fixed effects (Compare three or more groups)

One-way ANOVA, effect size f .

☐ Chi-square test (Goodness of fit or independence)

Goodness of fit or independence; effect size w .

☐ Pearson correlation test (Is a correlation different from zero?)

Test whether a correlation differs from zero.

Smallest expected path coefficient [β]

0.2

Significance level α

0.05

Power ($1 - \beta$)

0.8

Plan for a power of .80 at $\alpha = .05$ unless the field

Figure 24. The inverse square root method for the minimum sample size of a PLS-SEM model.

What it means for our example

To detect a medium effect ($f^2 = 0.15$) with four predictors, $\alpha = .05$ and power .80, you need $N = 85$ respondents. The inverse square root method (Kock & Hadaya, 2018) for a PLS-SEM model whose smallest expected path is .20 gives $N = 155$ (Figure 24). An independent-samples t-test needs 128 participants (64 per group) to detect $d = 0.50$.

Cut-off values: power

Plan for a power of .80 at $\alpha = .05$ unless your field expects otherwise; Cohen's (1988) conventions for small, medium and large effects are shown next to each effect size.

7. Machine learning

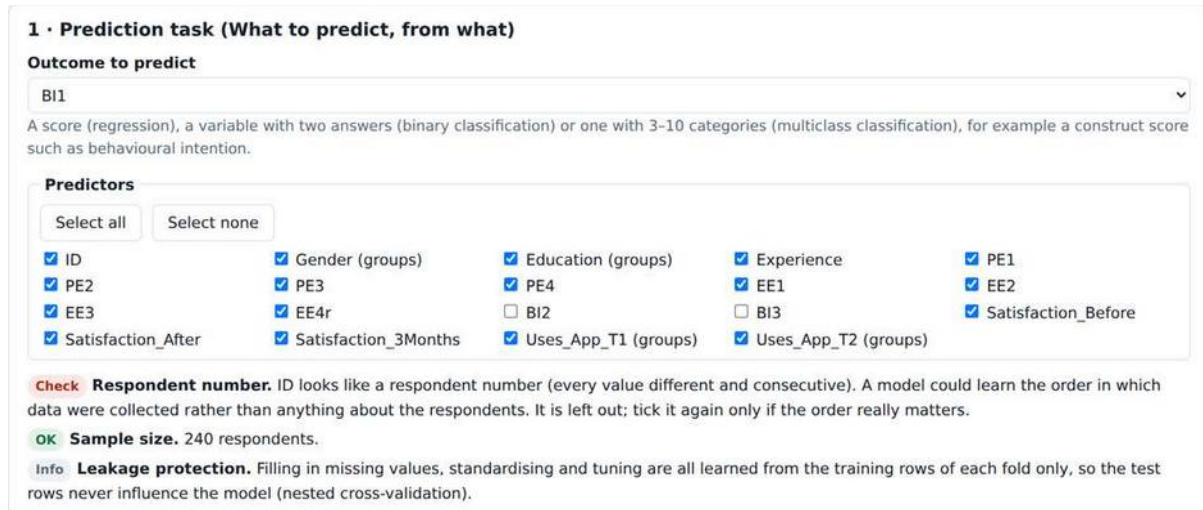
Concept: prediction and explanation

A regression or PLS-SEM model is judged by how well it explains the data it was estimated on. Machine learning asks a different question: how well does a model predict respondents it has never seen? SaNaML compares simple and flexible models on new respondents. If a flexible model (a random forest or boosted trees) predicts much better than a linear one, the relationships are probably curved or involve interactions; if it does not, a linear model such as PLS-SEM is adequate. It then explains the best model: which predictors drive the prediction, in which direction, and how.

Click **Machine learning** in the menu. The screen has three set-up cards and, after the run, the results.

7.1 Step 1 • The prediction task

1. In **Outcome to predict**, choose **BI1** (Figure 25). SaNaML sets the task (here a regression, because BI1 is a score; a two-level variable gives a yes/no classification and one with 3–10 categories a multiclass classification) and ticks all other variables as predictors.
2. Untick predictors that should not be used. In the example we untick BI2 and BI3, the other items of the same construct.
3. Read the checks below the predictors.



1 • Prediction task (What to predict, from what)

Outcome to predict

BI1

A score (regression), a variable with two answers (binary classification) or one with 3–10 categories (multiclass classification), for example a construct score such as behavioural intention.

Predictors

Select all Select none

☒ ID	☒ Gender (groups)	☒ Education (groups)	☒ Experience	☒ PE1
☒ PE2	☒ PE3	☒ PE4	☒ EE1	☒ EE2
☒ EE3	☒ EE4r	☐ BI2	☐ BI3	☒ Satisfaction_Before
☒ Satisfaction_After	☒ Satisfaction_3Months	☒ Uses_App_T1 (groups)	☒ Uses_App_T2 (groups)	

Check **Respondent number.** ID looks like a respondent number (every value different and consecutive). A model could learn the order in which data were collected rather than anything about the respondents. It is left out; tick it again only if the order really matters.

OK **Sample size.** 240 respondents.

Info **Leakage protection.** Filling in missing values, standardising and tuning are all learned from the training rows of each fold only, so the test rows never influence the model (nested cross-validation).

Figure 25. 1 • Prediction task: the outcome, the predictors and the checks.

Note: leakage

A predictor that contains the outcome, is part of it, or was measured after it makes the prediction look far better than it is (*leakage*). SaNaML flags predictors that are almost identical to the outcome ($|r| \geq .95$), variables that look like respondent numbers (every value different and consecutive; here ID, which is left out automatically), variables with many missing values or no variation, and rare categories. It also learns every preparation step (filling in missing values, standardising, tuning) inside each training set only.

7.2 Step 2 • The validation plan

Choose how the models are tested (Figure 26):

- **Repeated k-fold cross-validation** (recommended; for 240 respondents 10 folds, 5 repeats) for your reported results;
- **5-fold cross-validation, no repeats (Quick look)** to try things out;
- **10-fold cross-validation, 10 repeats (Thorough)** for the most stable estimate.

Main measure is RMSE for scores (or MAE, R^2 , Q^2_{predict}), AUC for yes/no outcomes and macro AUC for categories. The **Random seed** (2026) makes every split reproducible. Every model is tuned by an inner 5-fold cross-validation inside each training set (*nested cross-validation*).

2 • Validation plan (How the models are tested)

Cross-validation

☐ **Repeated 10-fold cross-validation, 5 repeats (Recommended for 240 respondents)**
Each repeat splits the respondents differently; averaging the repeats makes the comparison stable.

☒ **5-fold cross-validation, no repeats (Quick look)**
Fast but noisy; use it to try things out, not to report.

☐ **10-fold cross-validation, 10 repeats (Thorough)**
The most stable estimate; takes longest.

Main measure Random seed

Tuning uses an inner 5-fold cross-validation inside every training set (nested cross-validation). The seed makes every split reproducible; it is reported with the results.

Figure 26. 2 • Validation plan. For this manual we use the Quick look.

Concept: cross-validation

The respondents are split into k parts (*folds*). Each model is trained on $k - 1$ parts and tested on the remaining one, k times over, so every respondent is predicted once by a model that did not see them. Repeating the split several times averages out the luck of one particular split (Kuhn & Johnson, 2013).

7.3 Step 3 • The models

Tick the models to compare (Figure 27). The null model (always the average) is always included as the yardstick.

Table 4

The models of SaNaML

Model	What it is
Null model	Predicts the average for everyone. Any useful model must beat it.
Linear / logistic regression	The usual straight-line model. Nothing to tune.
Elastic net regression	A penalised linear model (lasso, ridge or a mix) that shrinks weak effects towards zero.
CART decision tree	A single tree of if-then splits; easy to read, but often less accurate.

Model	What it is
Random forest	500 trees grown on resampled data and averaged; captures non-linear effects and interactions.
XGBoost	Gradient boosting: many small trees, each correcting the last; usually among the most accurate for survey data.
LightGBM	Fast gradient boosting with leaf-wise trees.
Stacked ensemble (super learner)	A weighted mix of the other models (off by default; slower).

3 · Models (Which methods to compare)

☐ **Null model (No predictors: always the average)**

Predicts the average for everyone; any useful model must beat it. Nothing to tune.

☒ **Linear / logistic regression (Baseline)**

The usual straight-line model (linear regression for a score, logistic regression for a yes/no outcome). Nothing to tune.

☒ **Elastic net regression (Penalised linear model: lasso, ridge or a mix)**

A linear model that shrinks weak effects towards zero, which protects against overfitting with many predictors. Mix of lasso and ridge (0, 0.5, 1) and penalty strength (20 values), by inner cross-validation.

☒ **Classification and regression tree, CART (Decision tree)**

A single tree of if-then splits; easy to read but often less accurate. Tree depth (2 to 8) and minimum leaf size (5 or 15), by inner cross-validation.

☒ **Random forest (Many decision trees averaged)**

Hundreds of trees grown on resampled data and averaged; robust and captures non-linear effects and interactions. 500 trees; standard settings (variables tried per split: one third for scores, square root for yes/no).

☒ **XGBoost gradient boosting (Many small trees, each correcting the last)**

Trees added one after another, each fixing the errors of the previous ones; usually among the most accurate for survey-type data. Tree depth (2, 3, 4) and number of rounds (100, 250, 500), learning rate 0.05, by inner cross-validation.

☒ **LightGBM gradient boosting (Fast boosting with leaf-wise trees)**

Boosting that grows each tree leaf by leaf on binned values; fast, and often as accurate as XGBoost. Number of leaves (4, 8, 16) and number of rounds (100, 250, 500), learning rate 0.05, at least 10 respondents per leaf, by inner cross-validation.

☐ **Stacked ensemble, super learner (A weighted mix of the other models)**

Combines linear or logistic regression, random forest, XGBoost and LightGBM with weights learned from their out-of-fold predictions; often at least as good as the best of them. Slower. Weights by non-negative least squares on 5-fold out-of-fold predictions; base models with standard settings (forest 300 trees; XGBoost depth 3, 250 rounds; LightGBM 8 leaves, 250 rounds).

Run the comparison 41 steps on 1 background worker.

Figure 27. 3 · Models: all models except the stacked ensemble are ticked.

1. Click **Run the comparison**. A progress bar shows the steps and the number of background workers (Figure 28); **Stop** cancels the run. With the Quick look the run takes about a minute.



Figure 28. The progress bar during the run.

7.4 Results • Model comparison

Machine learning model comparison (Predicting BI1)

What it means: Elastic net predicted BI1 best on respondents it had not seen (RMSE = 0.990, 95% CI [0.827, 1.153]; $R^2 = .128$). It was not significantly better than linear regression ($p = .174$): the relationships appear largely linear and additive, which supports a linear model such as PLS-SEM for this outcome. The simplest model that is not significantly worse than the best is linear regression; prefer it when interpretability matters.

Check Respondent number. ID looks like a respondent number (every value different and consecutive). A model could learn the order in which data were collected rather than anything about the respondents. It is left out; tick it again only if the order really matters.

OK Sample size. 240 respondents.

Info Leakage protection. Filling in missing values, standardising and tuning are all learned from the training rows of each fold only, so the test rows never influence the model (nested cross-validation).

Rule: Tune and evaluate in separate (nested) cross-validation loops; tuning on the test rows makes performance look better than it is (Varma & Simon (2006); Cawley & Talbot (2010)).

OK Validation. 5 train/test splits; the same splits for every model, so the comparison is paired.

Rule: With survey-sized samples, repeat k-fold cross-validation several times to stabilise the estimate (Kuhn & Johnson (2013)).

Info Comparison test. Differences between models are tested with the corrected resampled t-test, which allows for the overlap between training sets.

Rule: Compare models on the same folds with the corrected resampled t-test; the ordinary t-test is too liberal because the training sets overlap (Nadeau & Bengio (2003); Bouckaert & Frank (2004)).

Info Sample size and flexible models. Flexible models often need many more respondents than linear models before they pay off; with a larger sample the ranking could change.

Rule: Flexible models (random forests, boosting) need much larger samples than regression to reach stable performance (van der Ploeg et al. (2014)).

Out-of-sample performance (nested cross-validation)						
Model	RMSE M (SD)	95% CI	MAE	R^2	Q^2_{predict}	Δ vs best (p)
Elastic net	0.990 (0.088)	[0.827, 1.153]	0.793	.128	.158	best

Figure 29. The start of the model comparison report: what it means and the checks.

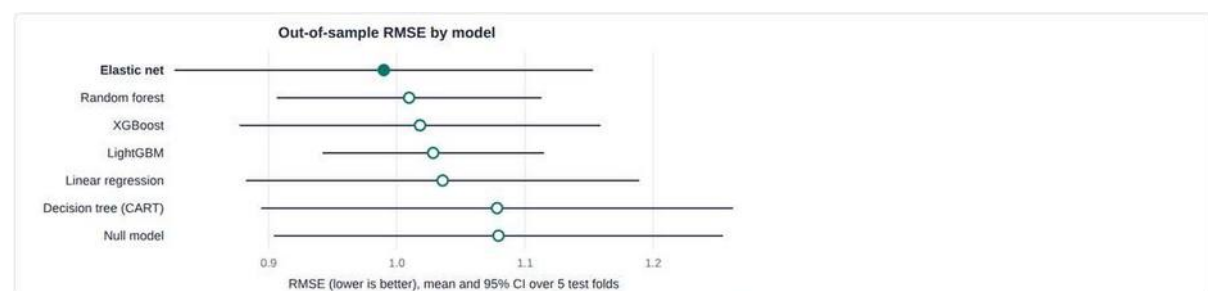


Figure 30. Out-of-sample RMSE of each model (dots: mean over the test folds; lines: 95% confidence intervals).

The report (Figure 29, Figure 30) ranks the models by their out-of-sample performance, with confidence intervals and a test of each model against the best (the corrected resampled t-test, which allows for the overlap between training sets; Nadeau & Bengio, 2003; Bouckaert & Frank, 2004). It also names *the simplest model that is not significantly worse than the best*.

Table 5

Out-of-sample prediction of BI1 (5-fold cross-validation)

Model	RMSE M (SD)	95% CI	R^2	Q^2_{predict}	Δ vs best (p)
Elastic net	0.990 (0.088)	[0.827, 1.153]	.128	.158	best
Random forest	1.010 (0.055)	[0.906, 1.113]	.088	.119	0.020 (.509)
XGBoost	1.018 (0.076)	[0.877, 1.159]	.073	.105	0.028 (.379)
LightGBM	1.028 (0.046)	[0.942, 1.115]	.052	.084	0.038 (.311)
Linear regression	1.036 (0.082)	[0.882, 1.189]	.041	.072	0.046 (.174)
Decision tree (CART)	1.078 (0.099)	[0.894, 1.262]	-.042	-.006	0.088 (.096)

Model	RMSE M (SD)	95% CI	R ²	Q ² predi ct	Δ vs best (p)
Null model	1.079 (0.094)	[0.904, 1.255]	-.035	.000	0.089 (.012)

What it means for our example

The elastic net predicts BI1 best (RMSE = 0.990), but not significantly better than linear regression ($p = .174$), and the flexible models are no better: the relationships appear largely linear and additive, which supports a linear model such as PLS-SEM for this outcome. Only the null model is significantly worse. The simplest model that is not significantly worse than the best is linear regression.

Note: small samples and flexible models

Flexible models need many more respondents than linear models before they pay off (van der Ploeg et al., 2014). With a few hundred respondents, "no better than linear" is a common and useful result. The Quick look is noisy: use repeated cross-validation for the results you report.

7.5 Results • Explanations

Explain this model chooses the model to explain; the options mark the best model and the simplest one that is not worse.

- **SHAP feature importance** (Figure 31): how much each predictor moves the prediction on average (mean $|\text{SHAP}|$, in points of the outcome), in which direction, and how its rank compares with the permutation importance on held-out folds. The SHAP summary plot (Figure 32) shows every respondent as a dot.
- **Accumulated local effects (ALE)** (Figure 33): how the prediction changes across the range of each main driver, holding the data realistic. A straight line means a linear effect; a bend or a plateau suggests a non-linear one. **Explore one predictor** draws the ALE with partial dependence (PDP) and individual curves (ICE) for any predictor.
- **The fitted decision tree** (Figure 34), when CART is explained.

SHAP feature importance (What drives the predictions of Elastic net)

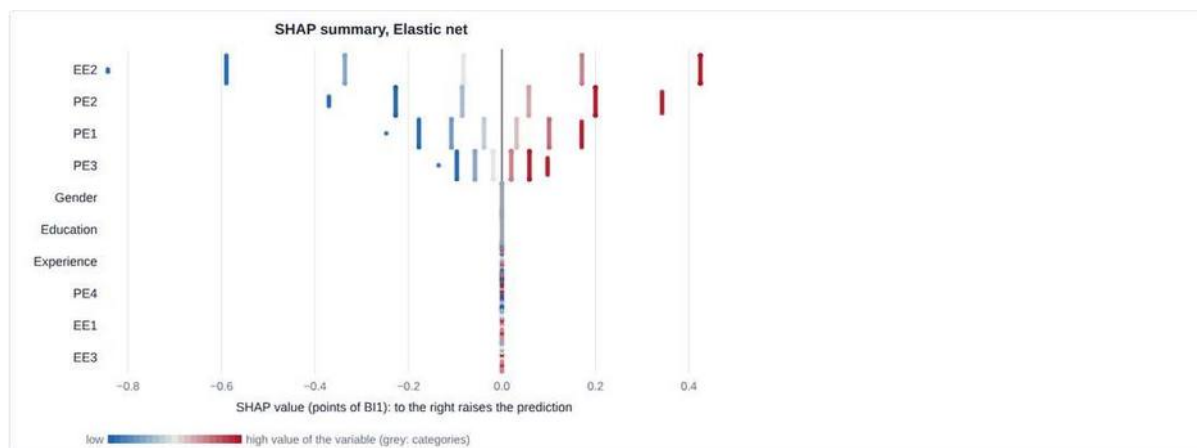
What it means: In the elastic net model, the strongest drivers of the prediction are EE2 (higher EE2 → higher predicted BI1), PE2 (higher PE2 → higher predicted BI1), PE1 (higher PE1 → higher predicted BI1). The SHAP ranking and the held-out permutation ranking differ noticeably; trust only drivers that rank high on both. Shuffling Gender, Education, EE4r, Satisfaction_Before and others did not hurt the predictions at all, so they add nothing beyond the other predictors in this model.

Info Meaning. Importance describes how this model uses the predictors; correlated predictors share credit, and importance is not evidence of a causal effect.

Rule: SHAP values, permutation importance, partial dependence and ALE describe what the model learned, not causal effects (Molnar (2022)).

Predictor	Mean SHAP	SHAP rank	Direction	Permutation importance M (SD)	Permutation rank
EE2	0.241	1	higher EE2 → higher predicted BI1	0.084 (0.039)	1
PE2	0.135	2	higher PE2 → higher predicted BI1	0.033 (0.015)	2
PE1	0.071	3	higher PE1 → higher predicted BI1	0.007 (0.010)	3
PE3	0.038	4	higher PE3 → higher predicted BI1	0.000 (0.004)	14
Gender	0.000	5	depends on the category	0.000 (0.000)	7
Education	0.000	6	depends on the category	0.000 (0.000)	8
Experience	0.000	7	mixed or non-linear	0.000 (0.001)	13
PE4	0.000	8	mixed or non-linear	0.000 (0.000)	6
EE1	0.000	9	mixed or non-linear	0.000 (0.001)	4
EE3	0.000	10	mixed or non-linear	0.000 (0.000)	5
EE4r	0.000	11	mixed or non-linear	0.000 (0.000)	9
Satisfaction_Before	0.000	12	mixed or non-linear	0.000 (0.000)	10
Satisfaction_After	0.000	13	mixed or non-linear	0.000 (0.000)	11
Satisfaction_3Months	0.000	14	mixed or non-linear	0.000 (0.001)	15

Figure 31. SHAP feature importance in the elastic net model.



SHAP summary plot (Elastic net). Each dot is one respondent. Position: how much that predictor moved this respondent's prediction; colour: the respondent's value on the predictor (red high, blue low).

Figure 32. SHAP summary plot: one dot per respondent and predictor.

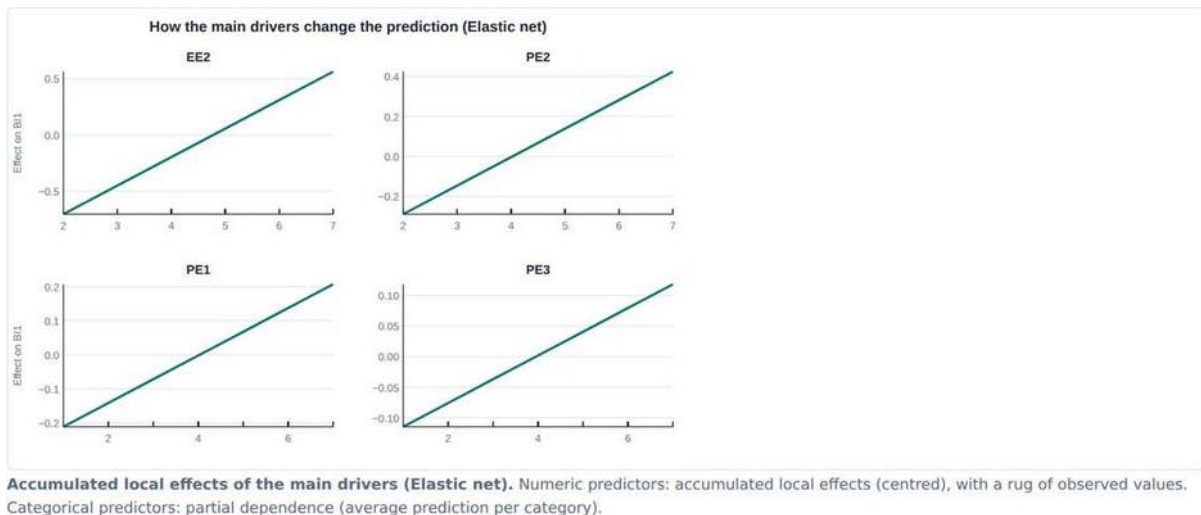


Figure 33. Accumulated local effects of the four main drivers.

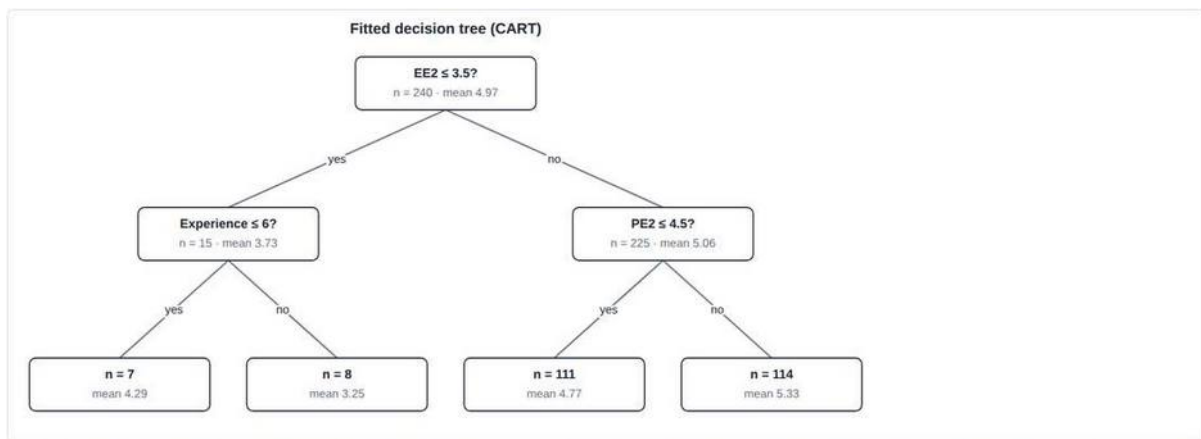


Figure 34. The fitted decision tree (CART), drawn as a diagram.

Cut-off values: reading explanations

SHAP values, permutation importance, partial dependence and ALE describe what the model learned, **not causal effects** (Molnar, 2022). Correlated predictors share credit. Trust a driver that ranks high on both SHAP and permutation importance. ALE is preferred over partial dependence when predictors are correlated, as survey items usually are (Apley & Zhu, 2020).

What it means for our example

EE2 (mean |SHAP| = 0.241), PE2 (0.135) and PE1 (0.071) drive the prediction most, all in the expected direction (higher answer → higher predicted BI1); the two rankings correlate at $p = .66$. Gender, Education and the satisfaction scores add nothing. The ALE curves are straight lines: across its observed range, EE2 raises the predicted BI1 by about 1.27 points, PE2 by about 0.71 and PE1 by about 0.42.

7.6 Results • Precision and calibration

- **Calibration** (Figure 35): do the predictions mean what they say? For scores, the actual values are regressed on the cross-validated predictions; a slope near 1 and a mean prediction error near 0 indicate good calibration (Van Calster et al., 2019). For yes/no outcomes, SaNaML reports the calibration slope and calibration-in-the-large with a reliability diagram.
- **Prediction intervals** (scores only): click [Compute prediction intervals for](#) SaNaML computes CV+ conformal intervals (Barber et al., 2021), a range that should contain a new respondent's actual value, and checks their coverage on unseen respondents (Figure 36).

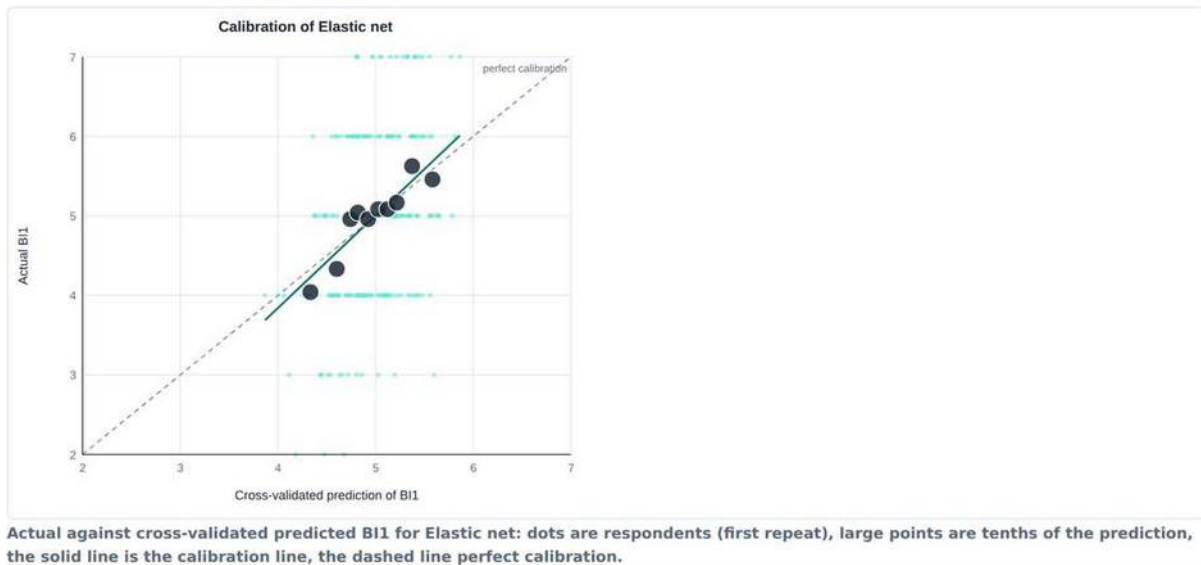


Figure 35. Calibration of the elastic net: actual against cross-validated predicted BI1.

Prediction intervals (How far off can one respondent's prediction be?) · XGBoost

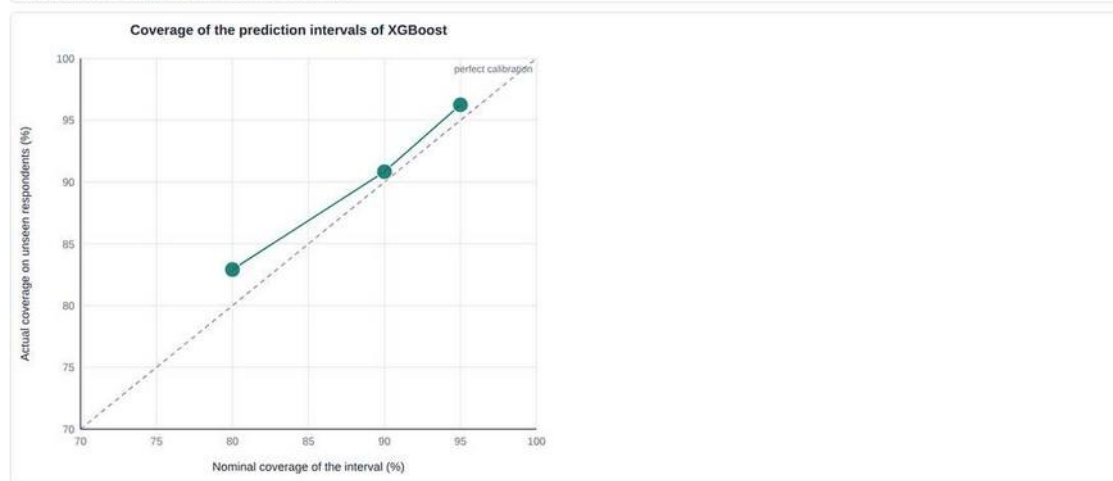
What it means: CV+ conformal prediction intervals (A range that should contain a new respondent's actual BI1) for XGBoost: a 90% interval was on average 3.47 wide on the BI1 scale (median 3.49) and contained the actual value for 90.8% of respondents the model had not seen. The intervals can be trusted at their stated level. Use the width to judge whether predictions for individuals are precise enough for your purpose; a model can rank respondents well and still be imprecise for any one person.

- OK 80% interval.** Contained the actual value for 82.9% of unseen respondents (target 80%, guaranteed at least 60%); median width 2.78.
Rule: A 90% prediction interval should contain about 90% of new respondents' values; CV+ intervals guarantee at least $1 - 2\alpha$ coverage if respondents are exchangeable and usually reach $1 - \alpha$ (Barber et al. (2021)).
- OK 90% interval.** Contained the actual value for 90.8% of unseen respondents (target 90%, guaranteed at least 80%); median width 3.49.
Rule: A 90% prediction interval should contain about 90% of new respondents' values; CV+ intervals guarantee at least $1 - 2\alpha$ coverage if respondents are exchangeable and usually reach $1 - \alpha$ (Barber et al. (2021)).
- OK 95% interval.** Contained the actual value for 96.3% of unseen respondents (target 95%, guaranteed at least 90%); median width 4.14.
Rule: A 90% prediction interval should contain about 90% of new respondents' values; CV+ intervals guarantee at least $1 - 2\alpha$ coverage if respondents are exchangeable and usually reach $1 - \alpha$ (Barber et al. (2021)).

Coverage and width of the prediction intervals of XGBoost on unseen respondents

Nominal level	Actual coverage	Guaranteed at least	Mean width	Median width
80%	82.9%	60%	2.77	2.78
90%	90.8%	80%	3.47	3.49
95%	96.3%	90%	4.22	4.14

Each of the 5 outer folds builds the intervals from its training respondents only (inner 5-fold CV+, tuned settings of the final model) and checks them on its test respondents; $n = 240$. Widths are in units of BI1.



Actual against nominal coverage of the prediction intervals of XGBoost: points on the dashed diagonal match their stated level.

How to report it (APA style): CV+ conformal prediction intervals (Barber et al., 2021) for XGBoost, evaluated in 5-fold cross-validation with an inner 5-fold CV+, achieved 82.9% coverage at the nominal 80% level (mean width 2.77), 90.8% coverage at the nominal 90% level (mean width 3.47), 96.3% coverage at the nominal 95% level (mean width 4.22).

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References

Figure 36. Prediction intervals for XGBoost: coverage and width at 80%, 90% and 95%.

What it means for our example

The elastic net is well calibrated (slope 1.16, 95% CI [0.81, 1.51]; mean error 0.001). For XGBoost, a 90% prediction interval is on average 3.47 points wide on the 1–7 scale and contained the actual value for 90.8% of unseen respondents: the intervals keep their promise, but a single respondent's intention cannot be predicted precisely. A model can rank respondents well and still be imprecise for any one person.

How to report it

"Seven prediction models were compared with 5-fold cross-validation (5 train/test splits), with every model tuned by an inner 5-fold cross-validation inside each training set (nested cross-validation; seed 2026). Elastic net performed best (RMSE = 0.990, 95% CI [0.827, 1.153]; R^2 = .128), but not significantly lower than linear regression, Δ RMSE = -0.046, corrected resampled $t(4) = -1.65$, $p = .174$."

8. The PLS-SEM link

The **PLS-SEM link** screen uses machine learning to test the structural model you estimated in SaNaPLS. It needs the scores and the model from a .sanahandoff file (section 3.4); without them it shows "No PLS-SEM model yet". In this chapter we use the UTAUT study: seven constructs, 303 respondents, with BI explained by PE, EE, SI, FC and HM, and UB by FC and BI.

8.1 Consistency check

The first card re-estimates every structural equation from the imported scores and compares the result with SaNaPLS (Figure 37).

PLS-SEM model from SaNaPLS (Consistency check)						
UTAUT demo · SaNaPLS 0.21.1 · 7 constructs · 7 paths · unstandardised scores.						
Structural equations reproduced from the imported scores						
Construct	Predictors	n	R ² (SaNaPLS)	R ² (scores)	Largest Δβ	Check
BI	PE, EE, SI, FC, HM	303	.497	.497	5.0e-7	OK The path coefficients into BI are reproduced from the scores (largest difference 5.0e-7).
UB	FC, BI	303	.305	.305	5.2e-8	OK The path coefficients into UB are reproduced from the scores (largest difference 5.2e-8).

In PLS-SEM the path coefficients are ordinary least squares coefficients of the standardised construct scores, so re-estimating them from the scores must give the SaNaPLS values. If it does, the scores and the model belong together.

Figure 37. PLS-SEM model from SaNaPLS: the consistency check.

What it means for our example

R² of BI (.497) and UB (.305) and all path coefficients are reproduced from the scores (largest difference 5.0e-7): the scores and the model belong together, and the following checks rest on the same model as your PLS-SEM results.

8.2 Benchmark and robustness checks

This card works on one structural equation at a time (Figure 38).

1. Choose the **Endogenous construct** (here **BI**, explained by PE, EE, SI, FC and HM).
2. Under **Predictors**, keep **Two-stage PLS-SEM → machine learning (Construct scores of the antecedents)**. The **Item-level benchmark** uses the retained items instead (benchmark only).
3. Choose the cross-validation (here **5-fold cross-validation, no repeats (Quick look)**) and the models.
4. Click **Run the benchmark and the robustness checks**.

Benchmark and robustness checks (One structural equation at a time)

Endogenous construct: BI (from PE, EE, SI, FC, HM) ▼

Predictors

- ☒ **Two-stage PLS-SEM → machine learning (Construct scores of the antecedents)**
The usual hybrid approach: the same inputs as the structural model. Includes the quadratic and interaction robustness checks.
- ☐ **Item-level benchmark (The retained items of the antecedents)**
Uses the items instead of the scores, so no full-sample weights enter the predictors. Benchmark only.

Cross-validation

- ☐ **Repeated k-fold cross-validation (Recommended for your sample size)**
Stable comparison; use this for reporting.
- ☒ **5-fold cross-validation, no repeats (Quick look)**
Fast and noisy.

Models

- ☒ Null model (No predictors: always the average)
- ☒ PLS-SEM structural equation (Linear regression on the scores)
- ☒ Elastic net regression (Penalised linear model: lasso, ridge or a mix)
- ☒ Random forest (Many decision trees averaged)
- ☒ XGBoost gradient boosting (Many small trees, each correcting the last)
- ☒ LightGBM gradient boosting (Fast boosting with leaf-wise trees)

Run the benchmark and the robustness checks

Figure 38. Benchmark and robustness checks: the settings.

Two reports appear:

- **PLS-SEM structural model benchmark** (Figure 39, Figure 40): can machine learning predict BI better than the PLS-SEM paths? It compares the out-of-sample Q^2_{predict} of the PLS-SEM equation (a linear regression on the construct scores) with the flexible models, and the path coefficients with the normalised SHAP importance.
- **Non-linear and interaction robustness checks** (Figure 41): tests every quadratic term and every two-way interaction of the antecedents with bootstrap intervals, Holm-adjusted p-values, f^2 , the cross-validated change in Q^2_{predict} , SHAP interaction values and Friedman's H^2 (Sarstedt et al., 2020).

PLS-SEM structural model benchmark (Can machine learning predict BI better than the PLS-SEM paths?)

What it means: Re-estimated in every training fold, the PLS-SEM structural equation of BI predicts new respondents with $Q^2_{\text{predict}} = .489$ (better than the training mean). The best non-linear model, LightGBM ($Q^2_{\text{predict}} = .440$), does significantly worse ($p = .036$): with these data the linear PLS-SEM paths predict BI better than a flexible model, so there is no sign of non-linear effects worth modelling. PE is the strongest predictor both by its path coefficient and in the lightGBM model.

Info Inputs. The construct scores come from the full-sample PLS-SEM model, so they are the same fixed inputs for every model; this comparison is fair between models, but its absolute Q^2_{predict} values are slightly optimistic.

Rule: Construct scores are fixed inputs from the full-sample PLS-SEM model; PLSpredict, which re-estimates the measurement model in every fold, is the stricter test of out-of-sample prediction (Shmueli et al. (2019)).

OK Benchmark. Q^2_{predict} of the linear equation = .489.

Rule: $Q^2_{\text{predict}} > 0$: the model predicts better than the training-sample mean (Shmueli et al. (2019)).

Info Comparison test. Model differences are tested on the same folds with the corrected resampled t-test.

Rule: Compare models on the same folds with the corrected resampled t-test; the ordinary t-test is too liberal because the training sets overlap (Nadeau & Bengio (2003); Bouckaert & Frank (2004)).

Out-of-sample prediction of BI					
Model	Q^2_{predict} M (SD)	95% CI	RMSE	MAE	Δ RMSE vs linear (p)
PLS-SEM structural equation (Linear regression on the construct scores)	.489 (.072)	[.355, .623]	0.789	0.640	-
Elastic net	.489 (.071)	[.357, .621]	0.789	0.641	0.000 (.616)
LightGBM	.440 (.059)	[.329, .550]	0.827	0.687	0.038 (.036)
XGBoost	.438 (.069)	[.310, .566]	0.828	0.684	0.039 (.030)
Random forest	.427 (.052)	[.330, .524]	0.837	0.698	0.048 (.024)

Figure 39. The start of the benchmark report.

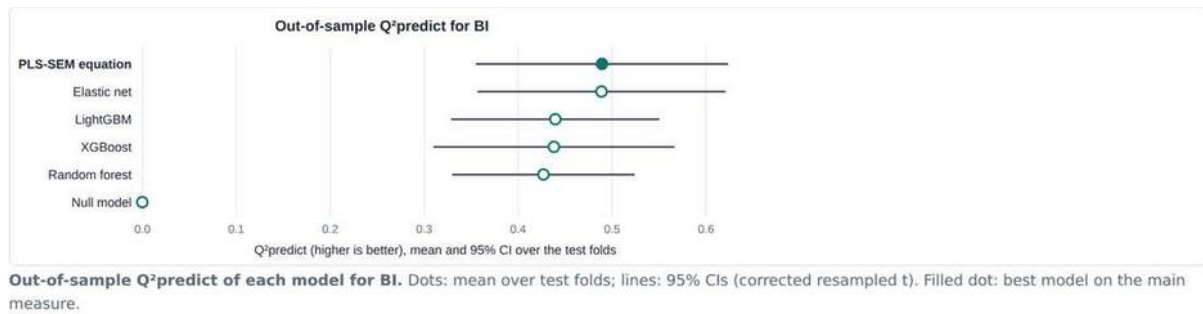


Figure 40. Out-of-sample Q^2 predict of each model for BI.

Non-linear and interaction robustness checks (Are there curved effects or interactions in the BI equation?)

What it means: No quadratic or interaction term is supported after Holm adjustment and cross-validation: the linear structural equation of BI is robust to these alternatives. In XGBoost the strongest interaction is PE $\times$ EE by SHAP interaction values and by the H-statistic ($H^2 = .012$, i.e. 1% of the joint effect of the pair).

Info Robustness checks. Quadratic effects and interactions are tested as alternatives to the linear paths; add a term only when it is both supported here and justified by theory.

Rule: Check the structural model for non-linear (quadratic) effects and add a term only when it is significant and theoretically justified (Sarstedt et al. (2020)).

Info Effect size. f^2 of the terms is judged against the smaller benchmarks used for interaction terms.

Rule: f^2 of an interaction or quadratic term: .005 small, .01 medium, .025 large (Kenny (2018)).

Info Multiple testing. 15 terms were tried, so p-values are Holm-adjusted.

Term	b	95% bootstrap CI	p	p (Holm)	f^2	ΔQ^2 predict (p)	SHAP interaction	H^2	Verdict
SI ²	0.059	[-0.013, 0.134]	.131	1.000	.008 (small)	0.002 (.585)	-	-	not supported
EE $\times$ SI	-0.044	[-0.111, 0.027]	.295	1.000	.004 (negligible)	-0.002 (.662)	0.021	.006	not supported
HM ²	0.037	[-0.035, 0.111]	.370	1.000	.003 (negligible)	0.000 (.886)	-	-	not supported
PE $\times$ HM	0.026	[-0.043, 0.098]	.546	1.000	.001 (negligible)	-0.001 (.185)	0.019	.002	not supported
EE $\times$ EC	0.025	[-0.049, 0.106]	.551	1.000	.001 (negligible)	-0.002 (.657)	0.008	.002	not supported

Figure 41. The start of the robustness report.

Cut-off values: robustness terms

Add a quadratic or interaction term only when it is supported here (Holm-adjusted $p < .05$, bootstrap interval excluding 0 and ΔQ^2 predict > 0) **and** justified by theory (Sarstedt et al., 2020). f^2 of such a term: .005 small, .01 medium, .025 large (Kenny, 2018).

What it means for our example

The PLS-SEM equation of BI predicts new respondents with Q^2 predict = .489; the best non-linear model, LightGBM (.440), is significantly worse ($p = .036$). PE is the strongest predictor both by its path coefficient and in LightGBM. None of the 15 quadratic and interaction terms is supported; the strongest interaction in XGBoost, PE $\times$ EE, accounts for about 1% of the pair's joint effect ($H^2 = .012$). The linear structural equation of BI is robust.

8.3 Further checks of the PLS-SEM model

The third card offers three more checks for the chosen construct (in the example we switch to **UB**):

- **Double machine learning path check** (Figure 42): each path is re-estimated while the other predictors may act non-linearly, modelled by a random forest, XGBoost, LightGBM or an elastic net (Chernozhukov et al., 2018). Click **Run the path check**.

- **Importance-performance map analysis (IPMA)** (Figure 43): which constructs matter most for the outcome (importance: total effect) and how well they perform (mean score on 0–100), classic and with XGBoost (Ringle & Sarstedt, 2016). Click [Run the importance-performance map](#); it needs unstandardised scores.
- **Screening of variables outside the model** (exploratory): would other variables in the file improve the prediction? The UTAUT file has none besides the items, the scores and the id, so the button is disabled.

Double machine learning path check (Do the paths into UB hold when the other predictors may act non-linearly?)

What it means: Every path into UB keeps its sign and significance when the other predictors are allowed to act non-linearly (double machine learning with random forest): the PLS-SEM conclusions for UB are robust to this kind of model misspecification.

Info Assumption. This checks robustness to non-linear effects of the measured predictors, not causality.

Rule: Double machine learning removes bias from flexible (non-linear) effects of the measured controls; it cannot correct for confounders that were not measured (Chernozhukov et al. (2018)).

Paths into UB: PLS-SEM and double machine learning

Path	PLS-SEM β (p)	DML θ (SE)	95% CI	p	β inside the CI	Same conclusion
FC → UB	0.264 (p < .001)	0.244 (0.052)	[0.143, 0.345]	< .001	yes	yes
BI → UB	0.434 (p < .001)	0.427 (0.046)	[0.337, 0.518]	< .001	yes	yes

θ : effect of the standardised predictor on the standardised UB, with the other predictors controlled flexibly by random forest; median over 5 repetitions of 5-fold cross-fitting (SE aggregated as in the DoubleML package). Same conclusion: same sign and same significance at .05.

How to report it (APA style): As a robustness check, each path into UB was re-estimated as the effect of the standardised predictor in a partially linear model, with the other predictors as controls modelled by random forest (double machine learning, partialling-out score, 5-fold cross-fitting repeated 5 times; Chernozhukov et al., 2018). FC → UB: $\theta = 0.244$, 95% CI [0.143, 0.345], $p < .001$ (PLS-SEM $\beta = 0.264$); BI → UB: $\theta = 0.427$, 95% CI [0.337, 0.518], $p < .001$ (PLS-SEM $\beta = 0.434$).

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Figure 42. Double machine learning path check for the paths into UB.

Importance-performance map analysis (Which constructs matter most for UB, and how well do they perform?)

What it means: BI is the most important driver of UB (a one-point gain in BI's 0-100 performance raises UB by 0.44 points). Improvement priority (above-average importance, below-average performance): FC. In XGBoost the importance of PE, EE, SI, HM differs from the linear estimate by more than a quarter: the effect may be non-linear, or less certain than the linear estimate suggests; look at the ALE plots in the Machine learning tab before acting on them.

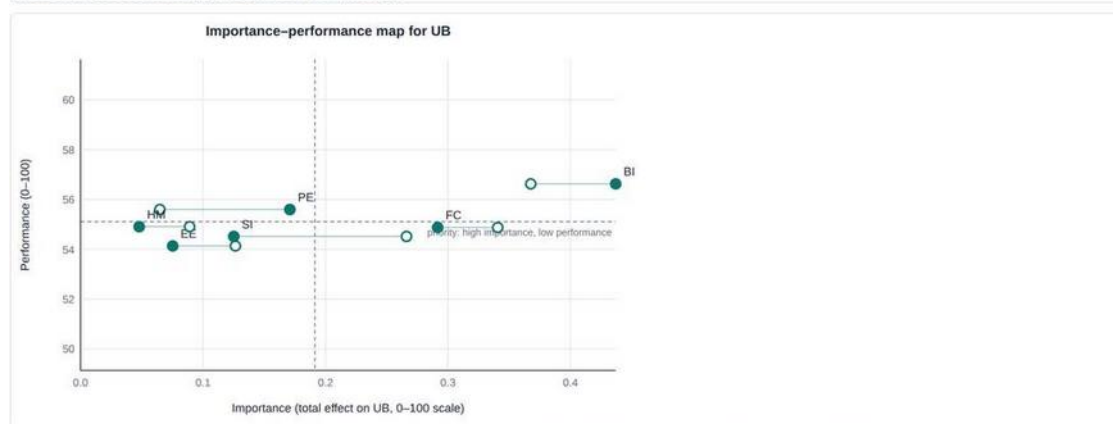
Info Priorities. Priority: FC.

Rule: Improvement priority: constructs with above-average importance and below-average performance (Ringle & Sarstedt (2016)).

Importance and performance for UB

Construct	Importance (PLS-SEM total effect)	Importance (linear, all antecedents)	Importance (XGBoost, ALE slope)	Performance (0-100)
PE	0.171	0.112	0.065	55.60
EE	0.075	0.048	0.126	54.13
SI	0.125	0.206	0.266	54.52
FC	0.292	0.299	0.341	54.88
HM	0.048	0.136	0.089	54.91
BI	0.437	0.406	0.368	56.63

Scale ranges used for the 0-100 rescaling: PE 1-5, EE 1-5, SI 1-5, FC 1-5, HM 1-5, BI 1-5, UB 1-7. PLS-SEM total effect: unstandardised, from the structural model (as SaNaPLS's IPMA). Linear and XGBoost: average slope of the accumulated local effect across the observed range, in a model of UB on the construct and the target's other antecedents that are not its consequences (differences from the first column show the effect of the model's restrictions; between the linear and XGBoost columns, the effect of non-linearity).



Importance-performance map for UB. Filled dots: PLS-SEM total effects; hollow dots: XGBoost ALE slopes; dashed lines: the means.

How to report it (APA style): An importance-performance map analysis (Ringle & Sarstedt, 2016) for UB ($n = 303$; scores rescaled to 0-100) showed PE (importance = 0.171, performance = 55.60), EE (importance = 0.075, performance = 54.13), SI (importance = 0.125, performance = 54.52), FC (importance = 0.292, performance = 54.88), HM (importance = 0.048, performance = 54.91), BI (importance = 0.437, performance = 56.63); the performance of UB was 54.13.

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Figure 43. Importance-performance map analysis for UB.

What it means for our example

Both paths into UB keep their sign and significance under double machine learning: $FC \rightarrow UB \theta = 0.244$ (PLS-SEM $\beta = 0.264$) and $BI \rightarrow UB \theta = 0.427$ ($\beta = 0.434$). In the importance-performance map, BI is the most important driver of UB (importance 0.437), and FC is the improvement priority: above-average importance (0.292) with below-average performance (54.88).

Note: what these checks can and cannot show

They test robustness to curved effects, interactions and flexible controls among the **measured** variables. They cannot correct for confounders that were not measured, and they do not turn a cross-sectional survey into causal evidence.

9. Heterogeneity: groups with different paths

The **Heterogeneity** screen looks for groups of respondents for whom the structural model works differently. It also needs the SaNaPLS model (Figure 44). We use *utaut_items_demo.csv*, which adds Gender, AgeGroup and Experience (novice or experienced).

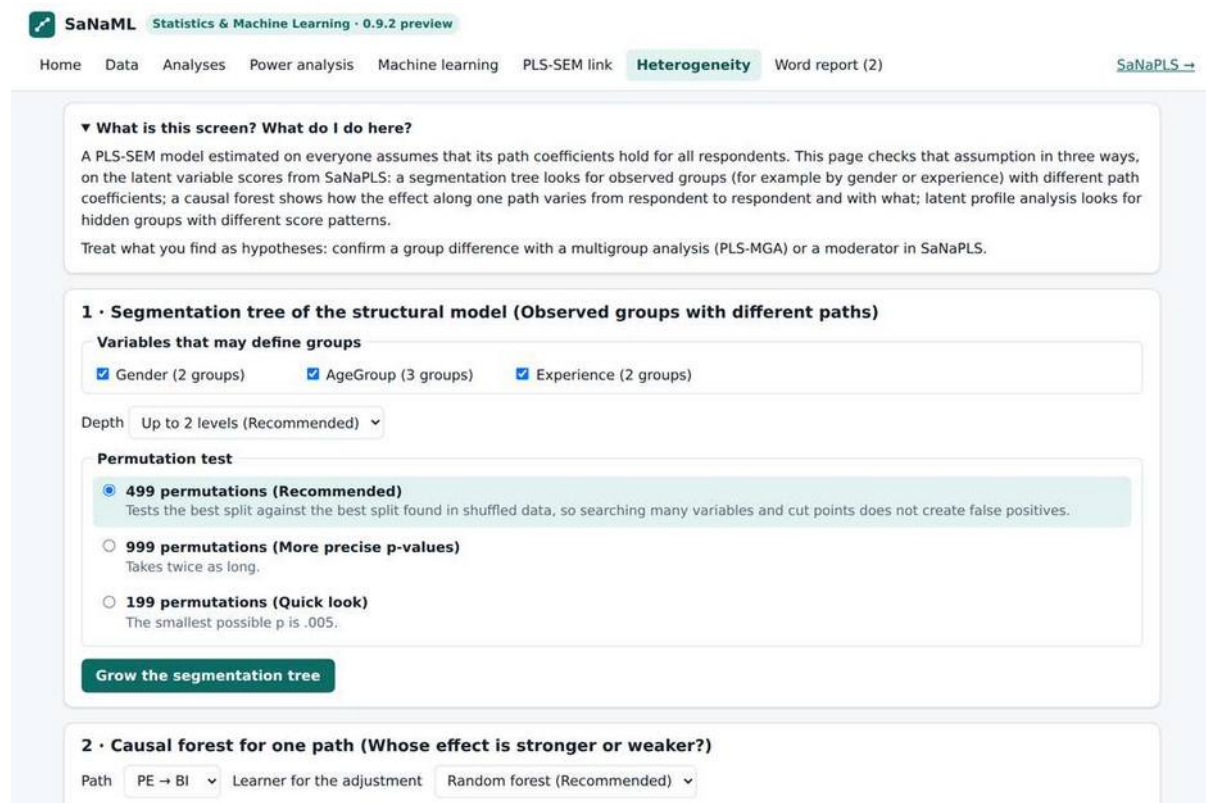


Figure 44. The Heterogeneity screen with its three analyses.

Note: hypotheses, not conclusions

Treat any group you find as a hypothesis. Confirm a group difference with a multigroup analysis (PLS-MGA) or a moderator in SaNaPLS, which also re-estimates the measurement model in each group and checks measurement invariance (MICOM).

9.1 Segmentation tree of the structural model

The segmentation tree (the PATHMOX approach; Lamberti et al., 2016) splits the respondents by the background variables whenever the split makes the path coefficients differ significantly.

1. Under **Variables that may define groups**, keep the variables ticked (Gender, AgeGroup, Experience).
2. Keep **Depth** at **Up to 2 levels (Recommended)** and choose the **Permutation test (499 permutations (Recommended))**; here **199 permutations (Quick look)**.
3. Click **Grow the segmentation tree** (Figure 45, Figure 46).

Segmentation tree of the structural model (Which groups of respondents have different path coefficients?)

What it means: The segmentation tree (PATHMOX approach) found groups of respondents with different path coefficients. The first split separates Experience = experienced from Experience = novice (global $F(7, 588) = 5.40$, permutation $p \leq .005$). The paths that differ most: PE → BI (0.61 vs -0.02, $p < .001$). Two segments result. Treat them as hypotheses to confirm with a multigroup analysis (PLS-MGA) in SaNaPLS, which also re-estimates the measurement model in each group.

OK Search-corrected test. Each split was tested with 199 permutations of the segmentation variables, which keeps the false-positive rate at 5% despite trying every variable and cut point.

Rule: Split only when the permutation test of the best split is significant ($p < .05$); the ordinary p of a split chosen from many candidates is far too small (Hothorn et al. (2006); Lamberti et al. (2016)).

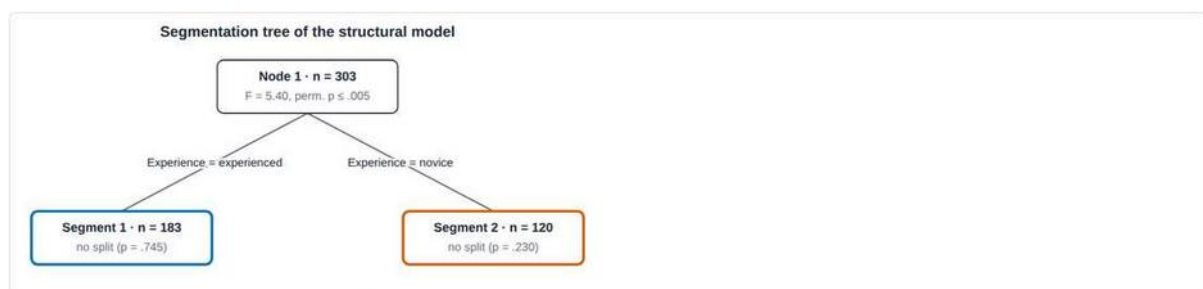
Info Segment size. Every group had at least 31 respondents (10% of 303); the tree grew to at most 2 levels.

Rule: Every segment needs enough respondents to estimate its equations: at least 10% of the sample (Lamberti et al. (2016)).

Info Fixed scores. Within segments, the paths are re-estimated from the full-sample construct scores. A PLS-SEM multigroup analysis re-estimates the measurement model in each segment and should confirm the result (after checking measurement invariance, MICOM).

Rule: Construct scores are fixed inputs from the full-sample PLS-SEM model; PLSpredict, which re-estimates the measurement model in every fold, is the stricter test of out-of-sample prediction (Shmueli et al. (2019)).

Figure 45. The start of the segmentation tree report.



Segmentation tree of the structural model: each split separates respondents whose path coefficients differ.

Figure 46. The segmentation tree: Experience separates two segments.

Cut-off values: segmentation tree

Split only when the permutation test of the best split is significant ($p < .05$); the ordinary p of a split chosen from many candidates is far too small (Hothorn et al., 2006; Lamberti et al., 2016). Every segment needs at least 10% of the sample.

What it means for our example

The first and only split separates experienced users ($n = 183$) from novices ($n = 120$), global $F(7, 588) = 5.40$, permutation $p \leq .005$. PE → BI differs most: 0.61 for experienced users against -0.02 for novices ($p < .001$). R^2 of BI is .654 among experienced users and .245 among novices.

9.2 Causal forest for one path

A causal forest (Athey et al., 2019; Wager & Athey, 2018) estimates how the effect of one path varies between respondents, after flexible adjustment for the other antecedents and the background variables.

1. Choose the **Path** (here PE → BI) and the **Learner for the adjustment** (Random forest (Recommended)).
2. Tick the moderators to examine (Gender, AgeGroup, Experience).
3. Click **Estimate how the effect of PE on BI varies**. It takes 10–60 seconds (Figure 47, Figure 48).

Causal forest (Whose effect is stronger or weaker?) · PE → BI

What it means: On average, a one-SD higher PE goes with a 0.44 SD higher BI (95% CI [0.35, 0.53]), close to the PLS-SEM path coefficient $\beta = 0.41$, after flexible adjustment for EE, SI, FC and HM (the other antecedents of BI) and for Gender, AgeGroup and Experience. The effect varies between respondents (calibration test: differential coefficient 1.10, $p < .001$): for the middle 80% of respondents it lies between 0.00 and 0.64. The variables that matter most are Experience, EE and FC. By Experience, the effect is 0.63 for experienced and -0.04 for novice. Consider testing Experience as a moderator of PE → BI in SaNaPLS.

Note Heterogeneity. The effect is not the same for everyone; report the average with the groups for which it differs.

Rule: Differential forest prediction significantly above 0: the effect varies with the moderators; near 1: the forest estimates that variation well (Chernozhukov et al. (2018); Athey et al. (2019)).

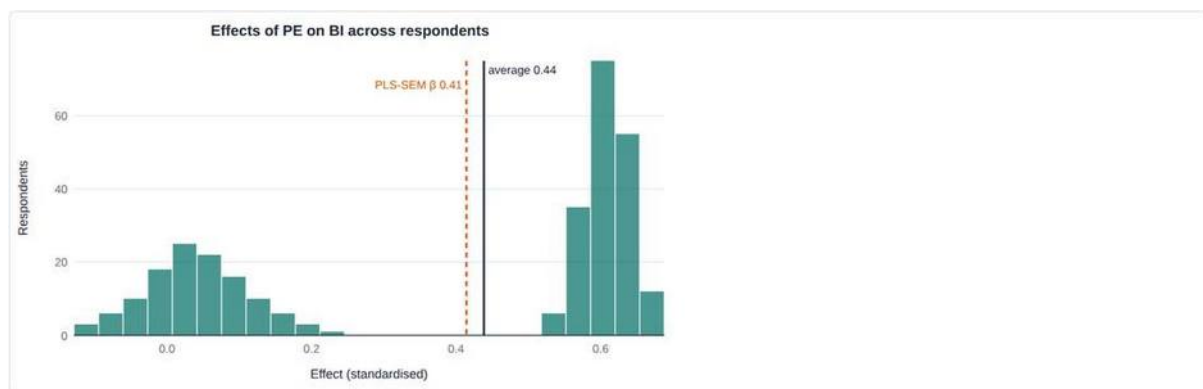
OK Honest estimation. Every tree chose its splits on one half of its sample and estimated the effects on the other half; the effects per respondent are out-of-bag (from trees that did not see the respondent).

Rule: Honest trees (splits chosen on one half, effects estimated on the other) give valid confidence intervals for the effects (Wager & Athey (2018)).

Info What it can and cannot show. The effects are conditional on the measured variables. Like the PLS-SEM paths, they are causal only if no unmeasured variable affects both PE and BI.

Rule: Double machine learning removes bias from flexible (non-linear) effects of the measured controls; it cannot correct for confounders that were not measured (Chernozhukov et al. (2018)).

Figure 47. The start of the causal forest report.



Distribution of the respondents' out-of-bag effects of PE on BI.

Figure 48. The distribution of the respondents' effects of PE on BI, with the average and the PLS-SEM path.

What it means for our example

On average, a one-SD higher PE goes with a 0.44 SD higher BI (95% CI [0.35, 0.53]), close to the PLS-SEM path ($\beta = 0.41$). The effect varies significantly between respondents (differential coefficient 1.10, $p < .001$), and the histogram has two humps. Experience matters most: the effect is 0.63 for experienced users and -0.04 for novices. The next step is to test Experience as a moderator of PE → BI in SaNaPLS.

9.3 Latent profile analysis

Latent profile analysis looks for hidden groups of respondents with different score profiles, and then compares the paths between the profiles.

1. Under **Constructs that define the profiles**, keep the antecedents (PE, EE, SI, FC, HM). The outcomes are marked "(outcome)" and are not ticked: forming groups on an outcome distorts its paths (Berk, 1983).
2. Click **Find latent profiles**. Solutions with 1 to 6 profiles are fitted, with 10 starts each.
3. Use **Show the solution with** to look at a solution (Figure 49, Figure 50).

Latent profile analysis (Are there hidden groups with different score profiles?)

What it means: The BIC favours 6 profiles (shown here: 2). Profile 1 (66%): close to the average on all. Profile 2 (34%): high PE, EE, HM. Relative entropy = .71 (profiles overlap; assignments of individual respondents are uncertain). The path coefficients do not differ significantly between the profiles (multigroup $F(7, 588) = 0.17, p = .991$).

Info Number of profiles. Chosen by the BIC, with every profile holding at least 5% of respondents; 10 starts per solution guard against local optima.

Rule: Choose the number of profiles by the BIC; every profile should hold at least 5% of respondents (Nylund et al. (2007); Spurk et al. (2020)).

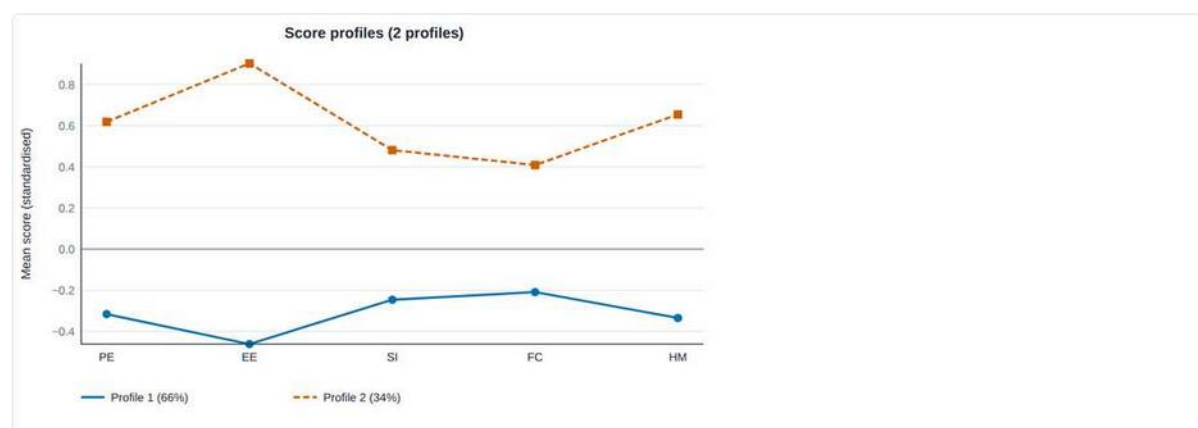
Note Separation. Relative entropy .71; lowest average posterior probability .88.

Rule: Relative entropy of .80 or more indicates clearly separated profiles (Spurk et al. (2020)).

Fit of solutions with 1 to 6 profiles

Profiles	Log-likelihood	AIC	BIC	SABIC	Entropy	Smallest profile	Lowest average posterior probability
1	-2147.2	4314.4	4351.5	4319.8	-	100.0%	-
2	-2076.3	4194.6	4272.5	4205.9	.71	33.7%	.88

Figure 49. The start of the latent profile report.



Profile plot: mean standardised scores of the 2 profiles.

Figure 50. Score profiles of the two-profile solution.

Cut-off values: latent profiles

Choose the number of profiles by the BIC, with every profile holding at least 5% of the respondents (Nylund et al., 2007). A relative entropy of .80 or more means the respondents are assigned clearly (Spurk et al., 2020).

What it means for our example

The BIC favours six profiles. The two-profile solution shown has a profile close to the average on everything (66%) and one high on PE, EE and HM (34%), with a relative entropy of .71 (the profiles overlap). The paths do not differ between the profiles (multigroup $F(7, 588) = 0.17, p = .991$): the difference found by the tree and the forest lies in how experience changes the effect of PE, not in the level of the scores.

10. The Word report

Every result has an **Add to Word report** button. After you click it, the button is replaced by "In the Word report" and a link **Open the report →**.

1. Click **Word report (5)** in the menu (Figure 51). The screen lists the results you added, each with its title and finding.
2. Change the order with $\uparrow$ and $\downarrow$, and take results out with **Remove**.
3. Type a **Report title** (default "Results").
4. Click **Download Word report (.docx)**. The file is named after the title, for example *Results (SaNaML).docx*.

The screenshot shows the SaNaML interface with a navigation bar at the top containing links for Home, Data, Analyses, Power analysis, Machine learning, PLS-SEM link, Heterogeneity, and Word report (5). The main content area is titled 'Word report (5 results)' and contains five entries, each with a title, a brief description, and controls for ordering and removal.

▼ What is this screen? What do I do here?
Collect the results you want to keep, then download them as one Word document: an APA-style sentence, the interpretation, the checks and numbered tables for each result, followed by the criteria used and the references. Add results with the **Add to Word report** button under any analysis.

Word report (5 results)

PLS-SEM structural model benchmark (Can machine learning predict BI better than the PLS-SEM paths?)
Re-estimated in every training fold, the PLS-SEM structural equation of BI predicts new respondents with $Q^2_{\text{predict}} = .489$ (better than the training mean). The best non-linear model, LightGBM ($Q^2_{\text{predict}} = .440$), does significantly worse ($p = .036$): with these data the linear PLS-SEM paths predict BI better than a flexible model, so there is no sign of non-linear effects worth modelling. PE is the strongest predictor both by its path coefficient and in the lightGBM model.

$\uparrow$ $\downarrow$ Remove

Non-linear and interaction robustness checks (Are there curved effects or interactions in the BI equation?)
No quadratic or interaction term is supported after Holm adjustment and cross-validation: the linear structural equation of BI is robust to these alternatives. In XGBoost the strongest interaction is PE $\times$ EE by SHAP interaction values and by the H-statistic ($H^2 = .012$, i.e. 1% of the joint effect of the pair).

$\uparrow$ $\downarrow$ Remove

Segmentation tree of the structural model (Which groups of respondents have different path coefficients?)
The segmentation tree (PATHMOX approach) found groups of respondents with different path coefficients. The first split separates Experience = experienced from Experience = novice (global $F(7, 588) = 5.40$, permutation $p \leq .005$). The paths that differ most: PE $\rightarrow$ BI (0.61 vs -0.02 , $p < .001$). Two segments result. Treat them as hypotheses to confirm with a multigroup analysis (PLS-MGA) in SaNaPLS, which also re-estimates the measurement model in each group.

$\uparrow$ $\downarrow$ Remove

Causal forest (Whose effect is stronger or weaker?) $\cdot$ PE $\rightarrow$ BI
On average, a one-SD higher PE goes with a 0.44 SD higher BI (95% CI [0.35, 0.53]), close to the PLS-SEM path coefficient $\beta = 0.41$, after flexible

Figure 51. The Word report screen with five results.

For each result the report contains the APA-style sentence, the interpretation, the checks and numbered tables, and the figures (as pictures). It ends with a table *Rules of thumb applied in this report* and the references of every source cited. Power analyses are not added to the report; copy their sentence instead.

Tip: keep your results

Results stay in the browser between visits, but they are lost if you clear the browser's site data or change computer. Download the Word report at the end of each working session.

Appendix A. Summary of cut-off values

Table 6

Rules of thumb used in SaNaML

Criterion	Rule of thumb	Source
Careless responding	Use several indices; exclude respondents flagged by 2 or more	Curran (2016); Meade & Craig (2012)
Univariate outliers	$ z > 3.29$ ($p < .001$)	Tabachnick & Fidell (2019)
Multivariate outliers	Mahalanobis D^2 with $p < .001$	Tabachnick & Fidell (2019)
Missing data	Under 10% per variable usually ignorable if random	Hair et al. (2019)
Little's MCAR test	$p \geq .05$: no evidence against MCAR	Little (1988)
Cronbach's α	$\geq .70$	Nunnally (1978)
Corrected item-total r	$\geq .30$	—
t-test	Welch's t-test by default	Delacre et al. (2017)
Cohen's d	0.2 / 0.5 / 0.8	Cohen (1988)
η^2 , ω^2	.01 / .06 / .14	Cohen (1988)
Unequal variances	Levene $p < .05$: Welch and Games-Howell	Levene (1960); Games & Howell (1976)
Sphericity	Greenhouse-Geisser if $\epsilon < .75$, Huynh-Feldt if $\epsilon \geq .75$	Girden (1992)
MANOVA	Report Pillai's trace	Olson (1976)
Expected counts	None below 1, $\leq 20\%$ below 5	Cochran (1954)
r , Cramér's V , ϕ	.10 / .30 / .50	Cohen (1988)
Regression sample size	$N \geq 50 + 8k$	Green (1991)
VIF	< 5 no concern; > 10 serious	Hair et al. (2019)
Logistic regression	EPV ≥ 10 ; AUC .7 / .8 / .9	Peduzzi et al. (1996); Hosmer et al. (2013)
Power	.80 at $\alpha = .05$	Cohen (1988)
PLS-SEM minimum sample	Inverse square root method	Kock & Hadaya (2018)
Model comparison	Corrected resampled t-test on the same folds	Nadeau & Bengio (2003); Bouckaert & Frank (2004)
Validation	Repeated k-fold, nested cross-validation	Kuhn & Johnson (2013)
Explanations	Describe the model, not causes	Molnar (2022); Apley & Zhu (2020)
Calibration	Slope close to 1, mean error close to 0	Van Calster et al. (2019)
Prediction intervals	Coverage close to the stated level; CV+ guarantees at least $1 - 2\alpha$	Barber et al. (2021)
Robustness terms	Supported and theoretically justified; f^2 .005 / .01 / .025	Sarstedt et al. (2020); Kenny (2018)
Double machine learning	Same sign and significance as PLS-SEM	Chernozhukov et al. (2018)

Criterion	Rule of thumb	Source
IPMA priority	Above-average importance, below-average performance	Ringle & Sarstedt (2016)
Segmentation tree	Permutation $p < .05$; each segment $\geq 10\%$	Hothorn et al. (2006); Lamberti et al. (2016)
Causal forest	Calibration test of heterogeneity	Athey et al. (2019); Chernozhukov et al. (2018)
Latent profiles	BIC, each profile $\geq 5\%$; entropy $\geq .80$	Nylund et al. (2007); Spurk et al. (2020)

Appendix B. Quick reference

Table 7

Where to find each analysis

To do this	Click
Open SaNaML	sanapls.com (or SaNaSoft.exe) → Statistics & Machine Learning →
Check the computer	Home → Check this computer
Open data	Data → Choose a file... (or Try the teaching data)
Build a construct score	Data → 3 · Build construct scores → name, quick pick, R for reversed items → Create the construct score
Bring scores from SaNaPLS	Data → 4 · Open a .sanahandoff file... → Add the scores
Screen for careless responders	Data → Check data quality → → Screen the data → Exclude the ... ticked respondents
Find the right test	Analyses → Test selection guide (Help me choose)
Descriptives, t-tests, ANOVA, chi-square, correlation, regression	Analyses → choose in the menu on the left
Sample size	Power analysis
Compare prediction models	Machine learning → outcome → validation plan → models → Run the comparison
Explain a model	Machine learning → Results · Explanations → Explain this model
Prediction intervals	Machine learning → Compute prediction intervals for ...
Test the PLS-SEM paths	PLS-SEM link → Run the benchmark and the robustness checks; Run the path check; Run the importance-performance map
Find groups with different paths	Heterogeneity → Grow the segmentation tree; Estimate how the effect ... varies; Find latent profiles
Word report	Add to Word report under each result → Word report → Download Word report (.docx)

Appendix C. Common questions and problems

Table 8

Troubleshooting

What you see	What to do
"No data yet"	Open a file or the teaching data on the Data screen first.
"No PLS-SEM model yet" on PLS-SEM link or Heterogeneity	Open your data and the .sanahandoff file from SaNaPLS on the Data screen (step 4) and click Add the scores.
A reverse-worded item lowers Cronbach's α or has a negative item-total r	Tick its R box in step 3.
"... looks like a respondent number" in machine learning	Nothing to do: the column is left out. Tick it again only if the row order really matters.
"Leakage" checks	Untick predictors that contain or are part of the outcome, or were measured after it.
Machine learning takes very long	Use the Quick look while you explore, untick slow models (stacked ensemble), and run the system check to see how fast the computer is.
The careless-response indices cannot be computed	They need at least four questionnaire items with whole-number answers.
My results are gone	Results are kept per browser. Open the same browser, and download the Word report at the end of each session.

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